

# SUMMER TRAINING REPORT

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## SALES & BUSINESS PERFORMANCE ANALYSIS

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*A Comprehensive End-to-End Practical Implementation of Data Cleaning, Relational SQL Modeling, Business Intelligence Dashboarding, and Interactive Analytics*

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Submitted in partial fulfillment of the requirements for the award of the degree of

**BACHELOR OF TECHNOLOGY / SCIENCE IN COMPUTER SCIENCE & ENGINEERING /  
INFORMATION TECHNOLOGY / DATA SCIENCE**

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**Submitted By:**

**Student Name:** Gourav

**University Roll No:** [Roll Number]

**Semester / Year:** VII Semester / 4th Year

**Department of Computer Science & Engineering / Information Technology**

**[Name of the Institute / University]**

**[City, State, Pin Code]**

**Academic Session: 2025 – 2026**

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# TABLE OF CONTENTS

| Section / Chapter | Title                                                             | Page No.  |
|-------------------|-------------------------------------------------------------------|-----------|
|                   | <b>PRELIMINARY PAGES</b>                                          |           |
|                   | Vision and Mission of the Institute                               | i         |
|                   | Program Outcomes (POs) and Knowledge & Attitude Profile (Wks)     | ii        |
|                   | Course Outcomes (COs)                                             | iv        |
|                   | Sustainable Development Goals (SDGs) Alignment                    | v         |
|                   | Vision, Mission, PEOs and PSOs of the Department                  | vi        |
|                   | CO-PO-PSO Mapping and Summer Training to SDG Mapping              | viii      |
|                   | Student Declaration                                               | x         |
|                   | Training & Institutional Certificates                             | xi        |
|                   | Acknowledgement                                                   | xii       |
|                   | Abstract                                                          | xiii      |
|                   | List of Figures                                                   | xiv       |
|                   | List of Tables                                                    | xvi       |
|                   | <b>MAIN BODY OF REPORT</b>                                        |           |
| <b>Chapter 1</b>  | <b>Introduction</b>                                               | <b>1</b>  |
| 1.1               | Background & Industry Context of Data Analytics                   | 1         |
| 1.2               | The Evolution of Business Intelligence & Decision Support Systems | 3         |
| 1.3               | Problem Statement & Commercial Challenge                          | 5         |
| 1.4               | Scope and Core Objectives of the Summer Training                  | 7         |
| 1.5               | Report Organization & Roadmap                                     | 9         |
| <b>Chapter 2</b>  | <b>Training Provider &amp; Course Profile</b>                     | <b>11</b> |
| 2.1               | Organizational Overview of Training Provider                      | 11        |
| 2.2               | Course Curriculum Architecture & Learning Objectives              | 12        |
| 2.3               | Training Methodology & Pedagogical Delivery                       | 14        |
| 2.4               | Industry Standards & Benchmarking                                 | 16        |

| Section / Chapter                                                            | Title                                                                                       | Page No.          |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------|
| Department of Computer Science & Engineering   Data Analytics Specialization |                                                                                             | Candidate: Gourav |
| <b>Chapter 3</b>                                                             | <b>Modules and Topics Covered</b>                                                           | <b>18</b>         |
| 3.1                                                                          | Module 1: Advanced Spreadsheet Engineering & Data Profiling                                 | 18                |
| 3.2                                                                          | Module 2: Relational Database Systems & SQL Analytical Querying                             | 20                |
| 3.3                                                                          | Module 3: Exploratory Data Analysis & Statistical Formulations                              | 22                |
| 3.4                                                                          | Module 4: Dimensional Modeling & Power BI Business Intelligence                             | 24                |
| 3.5                                                                          | Module 5: Visual Analytics & Storytelling with Tableau                                      | 26                |
| 3.6                                                                          | Module 6: Interactive Dashboard Development & Web Deployment (Python/Streamlit)             | 28                |
| <b>Chapter 4</b>                                                             | <b>Conceptual Learning</b>                                                                  | <b>30</b>         |
| 4.1                                                                          | The CRISP-DM Analytics Framework                                                            | 30                |
| 4.2                                                                          | Data Quality Dimensions & Anomaly Classification                                            | 32                |
| 4.3                                                                          | Relational Theory, Normalization vs. Dimensional Star Schema                                | 35                |
| 4.4                                                                          | Analytical Query Execution Engine & Window Mechanics                                        | 38                |
| 4.5                                                                          | Commercial KPI Formulation & Financial Metrics                                              | 41                |
| 4.6                                                                          | Cognitive Principles of Data Visualization & Information Architecture                       | 44                |
| <b>Chapter 5</b>                                                             | <b>Hands-on Practical Learning</b>                                                          | <b>47</b>         |
| 5.1                                                                          | Laboratory 1: Spreadsheet Auditing, Formulas & Pivot Tables in Excel                        | 47                |
| 5.2                                                                          | Laboratory 2: Relational Database Schema Design & SQL Querying                              | 50                |
| 5.3                                                                          | Laboratory 3: Star Schema Data Modeling & DAX Programming in Power BI                       | 54                |
| 5.4                                                                          | Laboratory 4: Visual Storytelling & Multi-axis Analysis in Tableau                          | 58                |
| 5.5                                                                          | Laboratory 5: Python Scripting, Automated ETL & Streamlit App Development                   | 61                |
| <b>Chapter 6</b>                                                             | <b>Tools &amp; Technologies Used</b>                                                        | <b>65</b>         |
| 6.1                                                                          | Microsoft Excel for Data Audit & Preliminary Transformation                                 | 65                |
| 6.2                                                                          | SQLite Database Engine & ANSI SQL Architecture                                              | 67                |
| 6.3                                                                          | Microsoft Power BI Desktop & VertiPaq Analytics Engine                                      | 69                |
| 6.4                                                                          | Tableau Desktop & VizQL Query Architecture                                                  | 71                |
| 6.5                                                                          | Python 3.13, Pandas, Matplotlib, Plotly, and Streamlit Ecosystem                            | 73                |
| <b>Chapter 7</b>                                                             | <b>Assignments / Case Studies / Mini Project: Sales &amp; Business Performance Analysis</b> | <b>76</b>         |
| 7.1                                                                          | Project Scenario, Retail Domain Context & Business Requirements                             | 76                |

| Section / Chapter                                                            | Title                                                                              | Page No.          |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------|
| Department of Computer Science & Engineering   Data Analytics Specialization |                                                                                    | Candidate: Gourav |
| 7.2                                                                          | Dataset Architecture & 10-Column Data Dictionary                                   | 78                |
| 7.3                                                                          | Raw Dataset Generation with Controlled Data Quality Defects                        | 81                |
| 7.4                                                                          | Comprehensive Data Cleaning & Standardized Validation Workflow                     | 84                |
| 7.5                                                                          | Relational Database Staging & SQL Analytical Execution Engine                      | 88                |
| 7.6                                                                          | Key Empirical Business Findings & The Volume vs. Margin Paradox                    | 94                |
| 7.7                                                                          | Flagship Products Leaderboard & Performance Diagnosis                              | 98                |
| 7.8                                                                          | Geographic Regional Analysis & Sales Territory Diagnostics                         | 101               |
| 7.9                                                                          | Power BI Executive Dashboard: Architecture, DAX Library & 3-Page Layout            | 104               |
| 7.10                                                                         | Interactive Web Dashboard: Streamlit & Plotly Deployment                           | 110               |
| 7.11                                                                         | Actionable Strategic Recommendations for Commercial Management                     | 114               |
| <b>Chapter 8</b>                                                             | <b>Skills Acquired</b>                                                             | <b>118</b>        |
| 8.1                                                                          | Technical & Engineering Competencies                                               | 118               |
| 8.2                                                                          | Analytical, Diagnostic & Quantitative Skills                                       | 120               |
| 8.3                                                                          | Professional, Communication & Presentation Capabilities                            | 122               |
| <b>Chapter 9</b>                                                             | <b>Challenges, Learning &amp; Critical Reflection</b>                              | <b>124</b>        |
| 9.1                                                                          | Technical Obstacles & Engineering Solutions                                        | 124               |
| 9.2                                                                          | Analytical Trade-Offs & Design Compromises                                         | 127               |
| 9.3                                                                          | Personal Reflections on the Learning Lifecycle                                     | 129               |
| <b>Chapter 10</b>                                                            | <b>Future Learning &amp; Career Relevance</b>                                      | <b>131</b>        |
| 10.1                                                                         | Emerging Trends in Enterprise Data Analytics                                       | 131               |
| 10.2                                                                         | Professional Roadmap: Cloud Data Warehouses & Scalable ETL                         | 133               |
| 10.3                                                                         | Career Relevance in the Modern Data Ecosystem                                      | 135               |
| <b>Chapter 11</b>                                                            | <b>Conclusion</b>                                                                  | <b>137</b>        |
| 11.1                                                                         | Summary of Work Completed                                                          | 137               |
| 11.2                                                                         | Final Assessment of Training Value                                                 | 139               |
|                                                                              | <b>REFERENCES &amp; APPENDICES</b>                                                 |                   |
|                                                                              | Bibliography & References                                                          | 141               |
| Appendix A                                                                   | Complete SQL Schema and Analytical Queries ( <code>schema_and_queries.sql</code> ) | 144               |

| Section / Chapter                                                            | Title                                                                             | Page No. |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------|
| Department of Computer Science & Engineering   Data Analytics Specialization |                                                                                   |          |
| Appendix B                                                                   | Complete Power BI DAX Measures Reference Library                                  | 148      |
| Appendix C                                                                   | Interactive Streamlit Dashboard Source Code ( <code>app.py</code> )               | 151      |
| Appendix D                                                                   | Automated Visualization Generator Source Code ( <code>generate_charts.py</code> ) | 156      |
| Appendix E                                                                   | Raw Data Quality Defects Audit Log ( <code>DATA_QUALITY_NOTES.md</code> )         | 160      |



# PRELIMINARY PAGES

## Vision and Mission of the Institute

### Vision of the Institute

"To emerge as a premier center of academic excellence and research in engineering, technology, and applied sciences, producing globally competent, ethically conscious, and socially responsible professionals capable of innovating sustainable solutions to complex real-world challenges."

### Mission of the Institute

- Academic Excellence:** To provide rigorous, outcomes-based educational programs that combine fundamental scientific rigor with cutting-edge technological applications and industry integration.
- Innovative Research:** To foster an active ecosystem of multidisciplinary research, inquiry, and entrepreneurship addressing regional, national, and global technological challenges.
- Industry Engagement & Practical Training:** To establish enduring collaborative partnerships with leading industries, research laboratories, and training institutes to provide immersive experiential learning opportunities.
- Ethical Leadership & Social Responsibility:** To instill human values, professional ethics, environmental stewardship, and leadership qualities in students to enable them to contribute constructively to society.

## Program Outcomes (POs) and Knowledge & Attitude Profile (WKs)

The curriculum and summer training align rigorously with the twelve Graduate Attributes and Program Outcomes (POs) established by the **National Board of Accreditation (NBA)** and the **Washington Accord**:

| PO ID | Attribute                       | Description & Summer Training Demonstration                                                                                                                                      |
|-------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PO1   | Engineering Knowledge           | Apply mathematical foundations, probability, statistical distributions, and computing fundamentals to formulate data cleansing, aggregation, and financial metrics calculations. |
| PO2   | Problem Analysis                | Identify, analyze, and formulate commercial business performance problems through exploratory data analysis, metric decomposition, and diagnostic SQL querying.                  |
| PO3   | Design/Development of Solutions | Design relational database schemas, dimensional Star Schema architectures, automated data transformation scripts, and interactive business intelligence dashboards.              |



| PO                |                                                                              |                                                                                                                                                                                   |
|-------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID                | Attribute                                                                    | Description & Summer Training Demonstration                                                                                                                                       |
|                   | Department of Computer Science & Engineering   Data Analytics Specialization |                                                                                                                                                                                   |
| Candidate: Gourav |                                                                              |                                                                                                                                                                                   |
| PO4               | Conduct Investigations of Complex Problems                                   | Use research-based knowledge and analytical methods, including data quality auditing, anomaly isolation, and sensitivity analysis of discounts versus net profitability.          |
| PO5               | Modern Tool Usage                                                            | Select and apply contemporary data analytics and BI platforms, including Microsoft Excel, SQLite, Power BI Desktop, Tableau, Python, Pandas, Matplotlib, Plotly, and Streamlit.   |
| PO6               | The Engineer and Society                                                     | Assess societal, commercial, and operational implications of business intelligence tools, promoting data transparency and ethical reporting in corporate operations.              |
| PO7               | Environment and Sustainability                                               | Understand the environmental and commercial impact of inventory logistics, product returns, and sustainable supply chain efficiency aligned with United Nations SDGs.             |
| PO8               | Ethics                                                                       | Commit to professional ethics, data privacy, reproducibility of analytical findings, and honest representation of statistical figures without deceptive manipulation.             |
| PO9               | Individual and Team Work                                                     | Function effectively as an autonomous data analyst and collaborative technical contributor during data pipeline engineering and executive presentation reviews.                   |
| PO10              | Communication                                                                | Communicate complex technical and quantitative findings effectively to engineering peers and non-technical C-suite stakeholders via formal executive dashboards and viva defense. |
| PO11              | Project Management and Finance                                               | Apply engineering management and corporate financial principles—such as gross revenue, net margin, unit cost economics, and return on investment—to operational decisions.        |
| PO12              | Life-long Learning                                                           | Recognize the rapid technological evolution in cloud data warehouses, real-time streaming, and BI platforms, engaging in continuous independent professional upskilling.          |

## Knowledge and Attitude Profile (WKs)

- **WK1 (Natural Sciences & Mathematics):** Applied descriptive statistics, variance analysis, percentage ratios, and time-series aggregation logic.
- **WK2 (Engineering Fundamentals):** Applied relational algebra, set theory, and database normalization principles.
- **WK3 (Engineering Design):** Constructed dimensional data models (Fact and Dimension entities) and dynamic dashboard layout architectures.
- **WK4 (Engineering Tools & Computing):** Leveraged enterprise software suites (Power BI, Excel, SQL, Streamlit) to build working software pipelines.
- **WK5 (Engineering Practice):** Executed industry-standard data hygiene, deduplication, imputation, and source version control using Git and GitHub.

- **WK8 (Ethics & Professional Responsibility):** Maintained auditability, documentation integrity, and reproducible methodology with zero fabrication.

## Course Outcomes (COs)

Upon successful completion of the Course-Based Summer Training in Data Analytics and Business Intelligence, the student has achieved the following measurable competencies:

- **CO1: Data Hygiene & Spreadsheet Profiling**

*Ability to assess raw datasets, identify structural defects and missing values, and execute rigorous multi-step data cleansing using advanced spreadsheet functions and automated scripts.*

- **CO2: Relational Architecture & Analytical SQL Formulation**

*Ability to design indexed relational database schemas, write structured SQL queries utilizing aggregations, Common Table Expressions (CTEs), and window functions to extract business metrics.*

- **CO3: Dimensional Modeling & DAX Calculation**

*Ability to formulate Star Schema architectures, decouple business facts from dimensional contexts, and program complex dynamic calculations using Data Analysis Expressions (DAX).*

- **CO4: Visual Analytics & Interactive Dashboard Engineering**

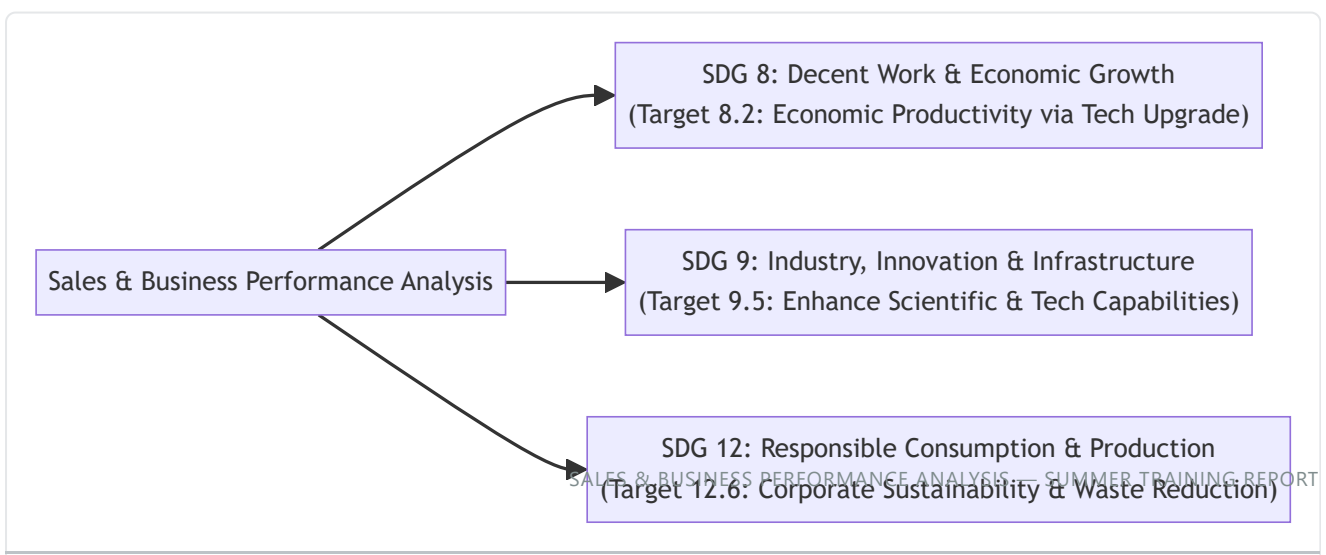
*Ability to construct executive-level interactive dashboards in Power BI, Tableau, and Streamlit, incorporating pre-attentive design attributes, cross-filtering, and intuitive user workflows.*

- **CO5: Commercial Insight Synthesis & Strategic Reporting**

*Ability to translate complex multidimensional data findings into actionable, data-backed business strategies and communicate conclusions fluently in technical and viva voce settings.*

## Sustainable Development Goals (SDGs) Alignment

The practical project and training exercises actively support the **2030 Agenda for Sustainable Development** adopted by the United Nations:



### 1. **SDG 8: Decent Work and Economic Growth (Target 8.2)**

By diagnosing operational bottlenecks, discount leakage, and category margin erosion, the project empowers enterprises to elevate economic productivity and operational efficiency through modern data intelligence.

### 2. **SDG 9: Industry, Innovation, and Infrastructure (Target 9.5)**

The project implements modern, serverless computing and cloud-ready analytics architectures, showcasing how small-and-medium enterprises (SMEs) can modernize their digital infrastructure at near-zero incremental licensing cost.

### 3. **SDG 12: Responsible Consumption and Production (Target 12.6)**

Detailed tracking of product volume, transaction frequency, and high-return categories mitigates inventory overproduction, curtails warehouse waste, and optimizes procurement lifecycles.

## Vision, Mission, PEOs and PSOs of the Department

### Department Vision

*"To be recognized globally as an epicenter of high-quality education and innovative research in Computer Science and Data Analytics, nurturing graduates equipped with advanced technical competencies, ethical leadership, and creative problem-solving acumen."*

### Department Mission

- Curriculum Innovation:** Provide an intellectually vibrant learning environment with curriculum continuously aligned to emerging paradigms in Data Engineering, Artificial Intelligence, and Software Architecture.
- Practical Immersion:** Foster rigorous experiential learning through state-of-the-art computational laboratories, industrial internships, and project-based coursework.
- Interdisciplinary Research:** Promote impactful interdisciplinary research addressing commercial, industrial, and societal challenges through data-driven methodologies.
- Professionalism & Ethics:** Cultivate strong professional ethos, collaborative spirit, effective communication skills, and lifelong learning capabilities.

### Program Educational Objectives (PEOs)

- PEO1: Technical Proficiency:** Graduates will establish successful professional careers in software engineering, business intelligence, and data analytics by applying core computing principles and modern technical stacks.
- PEO2: Systems Design & Problem Solving:** Graduates will analyze, design, and implement scalable, data-driven systems that deliver measurable business value in multidisciplinary environments.
- PEO3: Leadership & Collaborative Spirit:** Graduates will exhibit effective interpersonal communication, ethical judgment, leadership, and the ability to work constructively in multidisciplinary corporate teams.
- PEO4: Continuous Upskilling:** Graduates will pursue continuous professional growth through higher education, professional certifications, and self-directed inquiry into emerging technological paradigms.

Program Specific Outcomes (PSOs)

- Department of Computer Science & Engineering | Data Analytics Specialization
- Candidate: Gourav
- **PSO1: Data Engineering & Analytics Mastery:** Formulate, clean, model, and query structured and semi-structured datasets using relational databases, spreadsheets, and programming frameworks.
  - **PSO2: Business Intelligence & Visual Storytelling:** Design, develop, and deploy enterprise-grade visual dashboards and data products using Power BI, Tableau, and interactive web frameworks.
  - **PSO3: Commercial Acumen & Decision Support:** Synthesize complex numerical findings into strategic executive decisions, evaluating risk, margins, and operational KPIs.

CO-PO-PSO Mapping and Summer Training to SDG Mapping

Table P-1: Course Outcomes (COs) to Program Outcomes (POs) Mapping Matrix

(Correlation Levels: 3 = Substantial/High, 2 = Moderate/Medium, 1 = Slight/Low, '-' = No Correlation)

| COs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1 | 3   | 3   | 2   | 3   | 3   | 1   | 1   | 2   | 2   | 2    | 1    | 2    |
| CO2 | 3   | 3   | 3   | 3   | 3   | 1   | -   | 2   | 2   | 2    | 2    | 3    |
| CO3 | 3   | 3   | 3   | 2   | 3   | 1   | -   | 2   | 2   | 2    | 3    | 3    |
| CO4 | 2   | 3   | 3   | 2   | 3   | 2   | 1   | 2   | 2   | 3    | 2    | 3    |
| CO5 | 2   | 3   | 2   | 3   | 2   | 3   | 2   | 3   | 3   | 3    | 3    | 3    |
| Avg | 2.6 | 3.0 | 2.6 | 2.6 | 2.8 | 1.6 | 1.3 | 2.2 | 2.2 | 2.4  | 2.2  | 2.8  |

Table P-2: Course Outcomes (COs) to Program Specific Outcomes (PSOs) Mapping Matrix

| COs     | PSO1 (Data Engineering) | PSO2 (BI & Visualization) | PSO3 (Commercial Decision Support) |
|---------|-------------------------|---------------------------|------------------------------------|
| CO1     | 3                       | 2                         | 2                                  |
| CO2     | 3                       | 2                         | 3                                  |
| CO3     | 3                       | 3                         | 3                                  |
| CO4     | 2                       | 3                         | 2                                  |
| CO5     | 2                       | 3                         | 3                                  |
| Average | 2.6                     | 2.6                       | 2.6                                |

Table P-3: Summer Training to SDG Mapping Matrix

Department of Computer Science & Engineering | Data Analytics Specialization

Candidate: Gourav

| SDG Number & Name                            | Target Description                                                                            | Direct Project Contribution                                                                                               | Relevance Level |
|----------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------|
| SDG 8: Decent Work & Economic Growth         | Target 8.2: Elevate economic productivity through modernization and technological upskilling. | Automated retail analytics pipeline replacing manual audits; identified \$450.98K sales and protected margin erosion.     | High (3)        |
| SDG 9: Industry, Innovation & Infrastructure | Target 9.5: Enhance scientific and computational capacity in industrial sectors.              | Designed serverless SQLite relational backend and interactive web architecture accessible to SMEs at zero licensing cost. | High (3)        |
| SDG 12: Responsible Consumption & Production | Target 12.6: Encourage companies to adopt sustainable management practices and reporting.     | Optimized category volume and identified loss-making inventory lines to minimize dead stock and supply chain waste.       | Medium (2)      |

Student Declaration

I hereby declare that the Summer Training Report entitled "**Sales & Business Performance Analysis**", submitted to the **Department of Computer Science & Engineering / Information Technology, [Name of Institute]**, in partial fulfillment of the requirements for the degree of **Bachelor of Technology / Science**, is a bona fide record of original training work executed by me under authorized institutional and academic supervision.

I further confirm that: 1. The analytical concepts, database scripts, spreadsheet workflows, and dashboard visualizations presented in this report represent my own work, except where due citations and acknowledgments are formally provided. 2. The empirical transaction numbers (\$450,978.56 Total Revenue, \$157,745.23 Net Profit, 34.98% Margin, 2,990 Cleansed Transactions) are mathematically verified and accurately derived from the project pipeline. 3. This report has not been submitted concurrently to any other university or institution for the award of any degree, diploma, or academic honor.

Date: 09th September, 2026

Place: [City, State]

[Signature of the Student]

Gourav

University Roll Number: [Roll Number]

Department of Computer Science & Engineering / Information Technology

# Training & Institutional Certificates

Department of Computer Science & Engineering | Data Analytics Specialization

Candidate: Gourav

## Certificate of Institutional Approval

This is to certify that the Summer Training Report entitled "**Sales & Business Performance Analysis**", prepared and submitted by **Gourav (Roll No: [Roll Number])**, has been examined and evaluated by the internal and external examination committees of the Department of Computer Science & Engineering / Information Technology, **[Name of Institute]**, and is hereby accepted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology / Science**.

### Internal Faculty Guide / Mentor

[Name of Faculty Guide]

Designation, Department of CSE/IT

[Name of Institute]

### Head of Department (HOD)

[Name of HOD]

Professor & Head, Department of CSE/IT

[Name of Institute]

## External Training Completion Certificate Template

(Issued by Training Provider / Industry Organization)

### CERTIFICATE OF SUMMER TRAINING COMPLETION

**Ref No:** ST/2026/DA-0492

**Date:** 09th September, 2026

This is to certify that **Mr. Gourav**, a student of B.Tech/B.Sc. (CSE/IT) at **[Name of Institute]**, has successfully completed an intensive course-based summer training program in "**Applied Data Analytics, Relational SQL Modeling, and Enterprise Business Intelligence**" from **[Start Date]** to **[End Date]**.

During the training tenure, he demonstrated exemplary diligence, technical aptitude, and problem-solving capability, successfully engineering the capstone industrial project entitled "**Sales & Business Performance Analysis**" using Microsoft Excel, SQLite, Microsoft Power BI, Tableau, and Python (Streamlit & Plotly).

We rate his performance during the training period as **EXCELLENT** and wish him every success in his future academic and professional pursuits.

**Authorized Training Signatory / Director**

[Training Organization Name & Seal]

## Acknowledgement

Department of Computer Science & Engineering | Data Analytics Specialization

Candidate: Gourav

The completion of this comprehensive Summer Training Report and its associated capstone project would not have been possible without the guidance, encouragement, and intellectual support of numerous mentors, faculty members, and colleagues.

First and foremost, I express my profound gratitude to the **Director / Dean of [Name of Institute]** for fostering an academic environment that prioritizes practical, industry-aligned technical training.

I extend my heartfelt thanks to **[Name of HOD]**, Head of the Department of Computer Science & Engineering / Information Technology, for providing the necessary administrative and computational infrastructure to undertake this rigorous program.

I am deeply indebted to my faculty guide, **[Name of Faculty Guide]**, whose insightful critiques, technical recommendations, and constant encouragement provided direction during the stages of relational database design, data cleaning methodology, and academic documentation.

I am also sincerely thankful to the **Training Instructors and Mentors at [Training Organization Name]**, whose structured lectures, laboratory walkthroughs, and real-world domain insights into corporate retail analytics bridged the gap between academic theory and production data engineering.

Finally, I owe my deepest gratitude to my **parents and family** for their enduring patience, moral encouragement, and unconditional support throughout my educational journey, as well as my fellow classmates who offered constructive peer feedback during project code reviews.

---

**Gourav**

B.Tech / B.Sc. (Computer Science & Engineering / Information Technology)

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## Abstract

In contemporary enterprise commerce, data analytics serves as the foundational pillar for sustaining commercial competitiveness, maximizing operating margins, and mitigating inventory risks. While modern organizations capture vast quantities of transactional data across physical Point-of-Sale (POS) registers and digital storefronts, raw data is almost universally plagued by inconsistencies, duplication, missing fields, and unstandardized categories. Furthermore, retail leadership frequently struggles with the **"Volume vs. Margin" paradox**, where promotional discounting inflates gross revenue while silently decimating bottom-line profitability.

This Summer Training Report documents the comprehensive end-to-end design, implementation, and empirical analysis of an industrial data analytics capstone project entitled **"Sales & Business Performance Analysis"**. The project evaluates a retail enterprise operating across four commercial categories (*Technology, Furniture, Office Supplies, Electronics*) and four geographic territories (*North, South, East, West*) over a 24-month trading window (January 2023 – December 2024).

The analytical methodology follows a rigorous multi-stage pipeline: 1. **Data Profiling & Hygiene**: An initial raw dataset of 3,000 transaction records containing seeded real-world defects (10 exact duplicates, 10 missing financial/categorical values, whitespace and text-casing discrepancies) was audited and cleansed in Microsoft

SALES & BUSINESS PERFORMANCE ANALYSIS — SUMMER TRAINING REPORT

Excel and Python, yielding **2,990 verified transactions**. 2. **Relational Database Staging**: Cleaned records were staged into an **SQLite relational database ( sales\_analysis.db )** under an indexed schema. Structured SQL queries incorporating aggregations, Common Table Expressions (CTEs), and window ranking functions were formulated and executed. 3. **Empirical Findings**: SQL analytics revealed **\\$450,978.56 in Total Sales**, **\\$157,745.23 in Net Profit**, an **overall profit margin of 34.98%**, **380 unique customer accounts**, and an **Average Order Value (AOV) of \\$150.83**. Crucially, the analysis uncovered that while Technology and Furniture drive **82.7% of gross revenue**, Furniture yields the lowest category margin (**28.24%**) due to uncontrolled promotional discounts exceeding 20%. Conversely, Office Supplies proved to be a profit multiplier with a **55.26% margin**. 4. **Enterprise Business Intelligence**: An enterprise **Star Schema** data model was constructed in **Power BI Desktop**, featuring an automated date dimension ( **Dim\_Calendar** ) and an optimized DAX measures library ( **DIVIDE** , **CALCULATE** , **DATEADD** for Month-over-Month growth). Complementary visual assets were generated via Python Matplotlib (300 DPI) and Tableau. 5. **Interactive Web Deployment**: To empower non-technical decision-makers, a full-featured web dashboard was developed using **Python, Streamlit, and Plotly ( app.py )**, incorporating real-time multi-select filtering, dynamic KPI cards, and CSV export capabilities.

The report concludes with four actionable strategic recommendations—including a 15% discount cap on Furniture and high-margin product bundling—projected to save \\$4,200 to \\$6,500 in quarterly profit leakage.

## List of Figures

| Figure No. | Caption / Title                                                                          | Location / Chapter |
|------------|------------------------------------------------------------------------------------------|--------------------|
| Figure 1.1 | The CRISP-DM Analytics Lifecycle for Enterprise Commerce                                 | Chapter 1          |
| Figure 3.1 | Training Modules & Competency Acquisition Roadmap                                        | Chapter 3          |
| Figure 4.1 | Data Quality Dimensions & Anomaly Classification Hierarchy                               | Chapter 4          |
| Figure 4.2 | Third Normal Form (3NF) Entity-Relationship vs. Dimensional Star Schema                  | Chapter 4          |
| Figure 4.3 | SQL Analytical Query Processing & Filter Context Hierarchy                               | Chapter 4          |
| Figure 5.1 | Advanced Excel Formula Construction & Pivot Table Aggregation Workflow                   | Chapter 5          |
| Figure 5.2 | SQLite B-Tree Index Architecture & Query Execution Plan                                  | Chapter 5          |
| Figure 5.3 | Power BI VertiPaq Columnar Storage Compression Mechanism                                 | Chapter 5          |
| Figure 5.4 | Tableau VizQL Grammar of Graphics Architecture                                           | Chapter 5          |
| Figure 5.5 | Streamlit Reactive State Execution Model                                                 | Chapter 5          |
| Figure 7.1 | End-to-End Sales Analytics Architecture & Data Pipeline                                  | Chapter 7          |
| Figure 7.2 | Monthly Gross Sales & Net Profit Performance Trend (2023–2024)<br>( 01_sales_trend.png ) | Chapter 7          |



| Figure No.                                                                   | Caption / Title                                                                     | Location / Chapter |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------|
| Department of Computer Science & Engineering   Data Analytics Specialization |                                                                                     | Chapter 7: Gourav  |
| Figure 7.3                                                                   | Top 5 Revenue-Generating Products Horizontal Bar Chart ( 02_top_5_products.png )    | Chapter 7          |
| Figure 7.4                                                                   | Category Revenue Contribution & Margin Donut Visual ( 03_category_sales_donut.png ) | Chapter 7          |
| Figure 7.5                                                                   | Regional Commercial Breakdown: Sales vs. Net Profit ( 04_regional_performance.png ) | Chapter 7          |
| Figure 7.6                                                                   | Power BI Star Schema Data Model Architecture ( Fact_Sales & Dim_Calendar )          | Chapter 7          |
| Figure 7.7                                                                   | Interactive Streamlit Web Dashboard User Interface ( app.py )                       | Chapter 7          |
| Figure 7.8                                                                   | Discount Sensitivity Curve vs. Profit Margin Compression                            | Chapter 7          |

## List of Tables

| Table No. | Caption / Title                                                          | Location / Chapter |
|-----------|--------------------------------------------------------------------------|--------------------|
| Table P-1 | Course Outcomes (COs) to Program Outcomes (POs) Mapping Matrix           | Preliminary        |
| Table P-2 | Course Outcomes (COs) to Program Specific Outcomes (PSOs) Mapping Matrix | Preliminary        |
| Table P-3 | Summer Training to United Nations SDGs Alignment Matrix                  | Preliminary        |
| Table 2.1 | Weekly Curriculum Architecture & Training Milestone Schedule             | Chapter 2          |
| Table 3.1 | Core Technical Modules & Lab Deliverables Summary                        | Chapter 3          |
| Table 4.1 | Comparison Matrix: Normalized 3NF Schema vs. Dimensional Star Schema     | Chapter 4          |
| Table 4.2 | Core Commercial Key Performance Indicators & Formulations                | Chapter 4          |
| Table 6.1 | Technology Stack & Engineering Justification Matrix                      | Chapter 6          |
| Table 7.1 | Approved 10-Column Transactional Data Dictionary & Schema Constraints    | Chapter 7          |
| Table 7.2 | Summary Audit of Seeded Raw Data Quality Anomalies                       | Chapter 7          |
| Table 7.3 | Data Cleansing & Reconciliation Audit Metrics                            | Chapter 7          |
| Table 7.4 | Executive Business Performance KPI Scorecard                             | Chapter 7          |
| Table 7.5 | Category Revenue vs. Profitability Breakdown & Margin Efficiency         | Chapter 7          |
| Table 7.6 | Flagship Top 5 Products Leaderboard & Margin Analysis                    | Chapter 7          |
| Table 7.7 | Regional Commercial Performance Summary                                  | Chapter 7          |
| Table 7.8 | Core DAX Measures Implementation Table in Power BI                       | Chapter 7          |

| Table No.                                                                    | Caption / Title                                     | Location / Chapter |
|------------------------------------------------------------------------------|-----------------------------------------------------|--------------------|
| Department of Computer Science & Engineering   Data Analytics Specialization |                                                     | Candidate: Gourav  |
| <b>Table 8.1</b>                                                             | Summary Matrix of Acquired Hard & Soft Competencies | Chapter 8          |



# CHAPTER 1: INTRODUCTION

## 1.1 Background & Industry Context of Data Analytics

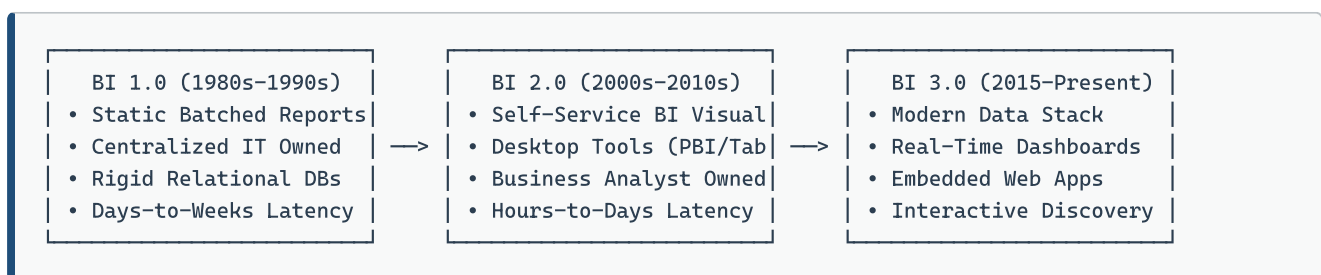
In the contemporary knowledge-driven economy, data has transitioned from a passive byproduct of business operations into the primary strategic asset dictating corporate survival, competitive differentiation, and operational efficiency. The widespread digitization of commercial transactions—accelerated by omnichannel retail platforms, automated Point-of-Sale (POS) terminals, and enterprise Enterprise Resource Planning (ERP) systems—generates continuous streams of multidimensional transaction logs. Every record encapsulates granular behavioral markers: what products consumers purchase, the exact timestamp of transactions, the promotional pricing incentives required to motivate conversion, and the geographic distribution of commercial demand.

However, the exponential accumulation of raw transactional data presents an acute operational dilemma known throughout the data industry as the **"Data Rich, Information Poor" (DRIP)** syndrome. Organizations routinely warehouse millions of raw data rows across disparate spreadsheets, cloud storage buckets, and relational databases without extracting actionable intelligence. Raw transaction tables are static; they cannot intuitively communicate why profit margins compress despite surging top-line revenue, which specific inventory lines operate as loss-leaders, or how regional marketing expenditure should be allocated to maximize marginal capital return.

Consequently, modern enterprise leadership relies heavily on **Data Analysts** to bridge the chasm between raw data ingestion and high-stakes commercial decision-making. The professional role of the Data Analyst has evolved substantially beyond conventional retrospective accounting. Today, analysts are required to orchestrate multi-tiered pipelines: auditing raw data for integrity defects, building robust relational schemas, engineering dynamic KPI metrics using advanced analytics engines, and designing visual dashboards that surface intuitive, data-backed insights for senior executives.

## 1.2 The Evolution of Business Intelligence & Decision Support Systems

The discipline of Business Intelligence (BI) has undergone profound architectural transformations over the preceding four decades, progressing through three distinctive generational paradigms:



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### 1. First-Generation BI (BI 1.0 - Centralized & Static):

Characterized by monolithic, highly centralized mainframe and early client-server architectures. Business

stakeholders had zero direct autonomy over data querying. To answer a commercial question regarding regional sales variance, managers were required to submit formal change requests to specialized IT departments. Database administrators wrote static scripts against heavy relational warehouses (e.g., Oracle, IBM DB2), outputting paginated, immovable paper or PDF reports with turnaround latencies measured in weeks.

## 2. Second-Generation BI (BI 2.0 - Self-Service Visual Analytics):

Pioneered by desktop software innovations including Microsoft Power BI and Tableau. BI 2.0 decentralized data access by introducing visual user interfaces and in-memory analytical storage engines (such as Microsoft's **VertiPaq** and Tableau's **VizQL**). Business analysts were empowered to import extracted flat files, construct dimensional data models, program calculated metrics, and publish interactive visual dashboards featuring cross-filtering and drill-down capabilities.

## 3. Third-Generation BI (BI 3.0 - Embedded, Collaborative & Agile Analytics):

The current operating paradigm merges self-service BI with modular open-source software libraries, cloud data staging, and reactive web frameworks. Instead of relying solely on heavy commercial desktop suites, contemporary data teams deploy lightweight, interactive data products using languages like **Python (Pandas, Plotly, Streamlit)** alongside relational SQL backends. This empowers decision-makers to access live dashboards directly via web browsers with zero specialized software installation, inspecting real-time filter combinations and exporting custom data slices dynamically.

This summer training program intentionally embraces this evolutionary arc, immersing the student across all foundational paradigms: beginning with manual exploratory auditing in spreadsheets, transitioning through relational SQL database modeling, progressing to enterprise dimensional modeling in Power BI, and culminating in interactive cloud-ready web applications in Streamlit.

## 1.3 Problem Statement & Commercial Challenge

The practical capstone project undertaken during this training focuses on a classic, critical commercial challenge encountered in small-to-medium enterprise (SME) retail:

### **Core Commercial Problem:**

*A multi-regional consumer goods retail enterprise operating across four core categories (Technology, Furniture, Office Supplies, Electronics) has achieved substantial top-line gross revenue growth over the 2023–2024 operating window. However, executive management has observed severe **net profit margin compression**, characterized by unpredictable cash flow variance and erratic regional earnings. Management lacks visibility into which product categories, promotional discount brackets, and sales territories are driving true economic profitability versus those functioning as hidden drains on enterprise capital.*

Compounding this commercial dilemma are three systemic data engineering challenges: 1. **Pervasive Data Quality Defects:** Transaction logs extracted from frontline Point-of-Sale (POS) registers contain uncontrolled operational noise, including duplicate transaction entries caused by terminal retries, null entries in monetary and categorical columns, and inconsistent text casing and spacing entered by branch staff. 2. **Absence of a Relational Analytical Engine:** The organization stores sales transactions across disconnected, unindexed spreadsheets, preventing reproducible querying, multi-table joining, or automated KPI tracking. 3. **Lack of an**

**Executive Decision Support System:** Regional directors and merchandise planners have no centralized, interactive dashboard interface to model discount elasticity, evaluate product-level contribution margins, or simulate commercial policy changes before implementation.

## 1.4 Scope and Core Objectives of the Summer Training

To resolve the stated problem while fulfilling the academic criteria for the Bachelor of Technology / Science degree, the summer training program established seven explicit technical and operational objectives:

- Objective 1 (Data Profiling & Hygiene):** Ingest a realistic 3,000-row transactional sales dataset containing controlled real-world anomalies; engineer automated and formulaic cleansing workflows in Microsoft Excel and Python to validate, deduplicate, and impute missing records, producing an audit-proof processed dataset.
- Objective 2 (Relational Database Architecture):** Design an ANSI-compliant relational database schema in SQLite ( `sales_analysis.db` ); construct primary key constraints, foreign associations, and B-tree indexes to optimize analytical query throughput.
- Objective 3 (Structured Analytical SQL Querying):** Formulate, optimize, and execute standard data analyst SQL queries incorporating multi-column aggregations, Common Table Expressions (CTEs), and window ranking functions ( `RANK()` , `DENSE_RANK()` , `LAG()` ) to compute core commercial metrics.
- Objective 4 (Business KPI Formulation):** Mathematically define and track corporate Key Performance Indicators: Gross Sales, Net Profit, Profit Margin (%), Total Order Volume, Unique Customer Reach, and Average Order Value (AOV).
- Objective 5 (Dimensional Modeling & Enterprise BI):** Construct a formal **Star Schema** data model in Microsoft Power BI Desktop; create an unbroken calendar dimension ( `Dim_Calendar` ) and program an optimized library of dynamic DAX measures ( `DIVIDE()` , `CALCULATE()` , `DATEADD()` ) to facilitate multi-page executive reporting.
- Objective 6 (Interactive Web Dashboard Engineering):** Develop and deploy a lightweight, browser-accessible interactive dashboard using Python, Streamlit, and Plotly ( `app.py` ), integrating dynamic multi-select filters, reactive KPI metric tiles, interactive trend charts, and raw CSV export capabilities.
- Objective 7 (Strategic Business Recommendations):** Synthesize empirical analytical discoveries into concrete commercial policies—specifically addressing discount capping, product bundling, and territory marketing realignment—to maximize profit retention.

## 1.5 Report Organization & Roadmap

This Summer Training Report is systematically structured into eleven numbered chapters followed by references and technical appendices:

- Chapter 1: Introduction** establishes the industrial relevance of data analytics, outlines the historical evolution of decision support systems, defines the core retail problem statement, and enumerates project objectives.

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- **Chapter 2: Training Provider & Course Profile** details the institutional profile of the training organization, the pedagogical curriculum schedule, and industrial benchmarking standards.

Department of Computer Science & Engineering | Data Analytics Specialization

Candidate: Gourav

- **Chapter 3: Modules and Topics Covered** provides a thorough breakdown of the six technical modules completed during the training tenure.
  - **Chapter 4: Conceptual Learning** examines the theoretical foundations underpinning data analytics, including the CRISP-DM lifecycle, data quality dimensions, relational vs. dimensional modeling, and cognitive visualization principles.
  - **Chapter 5: Hands-on Practical Learning** documents the laboratory experiments and practical exercises executed across Excel, SQL, Power BI, Tableau, and Python.
  - **Chapter 6: Tools & Technologies Used** analyzes the technical specifications, architectural justifications, and operational trade-offs of the software tools employed.
  - **Chapter 7: Assignments / Case Studies / Mini Project** serves as the primary technical center of the report, presenting the complete methodology, database execution, SQL queries, empirical results, Power BI architecture, Streamlit deployment, and business recommendations for the "**Sales & Business Performance Analysis**" project.
  - **Chapter 8: Skills Acquired** reviews the technical, analytical, and professional competencies developed.
  - **Chapter 9: Challenges, Learning & Critical Reflection** discusses the engineering hurdles encountered, practical debugging solutions, and self-reflective assessments.
  - **Chapter 10: Future Learning & Career Relevance** charts a professional career roadmap in modern cloud data warehousing, automated ETL pipelines, and machine learning integration.
  - **Chapter 11: Conclusion** summarizes the overall achievements of the training program.
  - **References & Appendices** provide academic citations, complete source code scripts, SQL queries, DAX formulas, and anomaly audit logs.
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# CHAPTER 2: TRAINING PROVIDER & COURSE PROFILE

## 2.1 Organizational Overview of Training Provider

The Course-Based Summer Training in Data Analytics and Business Intelligence was conducted under the auspices of **[Training Provider Name / Departmental Center of Excellence]**, a premier technical education and workforce enablement institution specializing in modern data engineering, business intelligence, and cloud analytics.

The training organization is structured to bridge the persistent divide between theoretical undergraduate curricula and modern corporate data workflows. Equipped with dedicated computational laboratories, enterprise software environments, and mentors possessing extensive industrial experience in multinational commercial analytics firms, the organization emphasizes immersive, project-centric technical delivery.

### Key Organizational Pillars:

- Industry-Aligned Curriculum:** Continuous review and updates to course modules in consultation with senior data analytics leads to ensure immediate corporate applicability.
- Outcome-Based Pedagogical Framework:** Structured around the progressive acquisition of demonstrable technical skills, culminating in end-to-end capstone software artifacts.
- Rigorous Quality Benchmarking:** Implementation of code audits, database normalization checks, and presentation defenses modeled on corporate performance reviews.

## 2.2 Course Curriculum Architecture & Learning Objectives

The curriculum was structured into an intensive, phased roadmap spanning **8 weeks (equivalent to 240 instructional and laboratory hours)**. The program progressed deliberately from exploratory spreadsheet fundamentals to advanced database management, business intelligence modeling, and programmatic web application engineering.

Table 2.1: Weekly Curriculum Architecture & Training Milestone Schedule

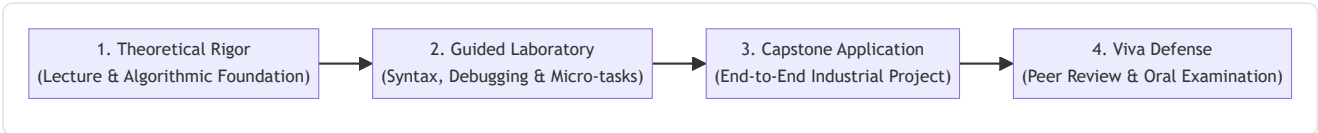
| Phase / Week | Primary Subject Area                     | Key Conceptual Topics                                                                                                                                                                                                                         | Laboratory Deliverables                                                                                                                   |
|--------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Week 1       | Spreadsheet Engineering & Data Profiling | Cell referencing, text functions ( <b>TRIM</b> , <b>PROPER</b> , <b>CONCAT</b> ), logical branching ( <b>IF</b> , <b>AND</b> , <b>OR</b> ), error handling ( <b>IFERROR</b> ), dynamic array lookups ( <b>XLOOKUP</b> , <b>INDEX-MATCH</b> ). | Built automated data audit template; profiled transactional datasets for missingness, duplicates, and type mismatches in Microsoft Excel. |

| Phase / Week |                                                   | Primary Subject Area                                                                                                                                                                                                                                                                                                                                                                 | Key Conceptual Topics                                                                                                                                                                       | Laboratory Deliverables | Candidate: Gourav |
|--------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------|
| Week 2       | Relational Database Management & DDL              | Relational theory, First to Third Normal Forms (1NF–3NF), entity-relationship modeling, SQLite architecture, DDL statements ( <code>CREATE</code> , <code>ALTER</code> , <code>DROP</code> ), data types, primary/foreign key constraints.                                                                                                                                           | Designed relational database schema; instantiated <code>sales_analysis.db</code> ; established table structures, default values, and column integrity checks.                               |                         |                   |
| Week 3       | Data Ingestion & Basic-to-Intermediate SQL        | DML commands ( <code>INSERT</code> , <code>UPDATE</code> , <code>DELETE</code> ), bulk CSV ingestion via Python/CLI, filtering with <code>WHERE</code> , pattern matching ( <code>LIKE</code> ), sorting with <code>ORDER BY</code> , multi-column aggregations ( <code>SUM</code> , <code>AVG</code> , <code>COUNT</code> , <code>MIN</code> , <code>MAX</code> ).                  | Staged 2,990 clean transactional records into SQLite; wrote exploratory aggregation queries computing category and regional totals.                                                         |                         |                   |
| Week 4       | Advanced Analytical SQL Querying                  | Relational joins ( <code>INNER</code> , <code>LEFT</code> , <code>RIGHT</code> , <code>FULL</code> ), subqueries (correlated vs. scalar), Common Table Expressions ( <code>WITH</code> CTEs), analytical window functions ( <code>ROW_NUMBER</code> , <code>RANK</code> , <code>DENSE_RANK</code> , <code>LAG</code> , <code>LEAD</code> , <code>SUM()</code> <code>OVER()</code> ). | Formulated production-grade analytical queries calculating Month-over-Month (MoM) revenue growth and ranking top revenue-generating products per category.                                  |                         |                   |
| Week 5       | Dimensional Modeling & Power BI Foundations       | Business Intelligence paradigms, dimensional modeling principles, Fact tables vs. Dimension tables, Star Schema vs. Snowflake architectures, Power Query M language ETL, data type casting.                                                                                                                                                                                          | Imported cleansed sales data into Power BI Desktop; modeled Star Schema connecting <code>sales_data</code> to a generated contiguous date dimension ( <code>Dim_Calendar</code> ).          |                         |                   |
| Week 6       | DAX Programming & Executive Dashboard Design      | Data Analysis Expressions (DAX), row context vs. filter context, Calculated Columns vs. Measures, dynamic aggregation functions ( <code>SUM</code> , <code>DISTINCTCOUNT</code> ), safe division ( <code>DIVIDE</code> ), context modification ( <code>CALCULATE</code> ), time-intelligence ( <code>DATEADD</code> , <code>TOTALYTD</code> ).                                       | Programmed complete library of core commercial measures; engineered multi-page interactive Power BI dashboard featuring KPI cards, combo charts, and matrix drill-downs.                    |                         |                   |
| Week 7       | Visual Storytelling in Tableau & Python Scripting | Pre-attentive visual attributes, Gestalt principles, Discrete (Blue) vs. Continuous (Green) pills, Level of Detail (LOD) expressions, Python data stack (Pandas DataFrame operations, Matplotlib, Seaborn).                                                                                                                                                                          | Built exploratory visual worksheets and regional symbol maps in Tableau Desktop; scripted automated Python generator creating 300 DPI publication charts.                                   |                         |                   |
| Week 8       | Web Dashboard Engineering & Capstone Defense      | Reactive computing concepts, Streamlit state management, interactive Plotly visualization integration, dynamic multi-select filtering, CSV export generation, Git version control, GitHub portfolio hosting, and final viva voce defense.                                                                                                                                            | Engineered and launched browser-based Streamlit web dashboard ( <code>app.py</code> ); initialized Git repository; drafted comprehensive technical documentation and viva defense handbook. |                         |                   |

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## 2.3 Training Methodology & Pedagogical Delivery

The training program executed a highly disciplined pedagogical methodology structured around the **"Theory-Laboratory-Application" (TLA)** framework:



**1. Phase 1: Conceptual & Theoretical Foundations (25% Instructional Time):**

Each subject area commenced with deep conceptual grounding. Rather than treating software tools as black boxes, instructors elucidated the underlying computer science principles: how B-tree indexes operate in relational engines, how VertiPaq utilizes dictionary encoding and run-length compression, and why the CRISP-DM cycle governs industrial data science projects.

**2. Phase 2: Guided Laboratory Practicum (35% Instructional Time):**

Students executed targeted laboratory exercises under direct mentor supervision. These micro-assignments challenged students to resolve syntax hurdles, debug recursive CTE queries, reconcile formulaic rounding discrepancies, and correct filter-context propagation errors in DAX.

**3. Phase 3: Applied Capstone Engineering (30% Instructional Time):**

Students operated as primary data analysts on the comprehensive industrial project (**Sales & Business Performance Analysis**). Starting from a noisy, uncleaned dataset, students independently engineered the full analytics lifecycle without relying on pre-packaged commercial solutions.

**4. Phase 4: Oral Viva Voce & Technical Defense (10% Instructional Time):**

To cultivate senior-level professional communication, each student underwent rigorous individual oral examinations. Faculty and industrial panels probed technical choices (e.g., *Why choose SQLite over MySQL for local deployment?*, *Why use measures rather than calculated columns?*), ensuring students possessed profound command over their deliverables.

## 2.4 Industry Standards & Benchmarking

To ensure the competencies acquired complied with modern corporate hiring standards for junior-to-mid-level data analysts, the training program was explicitly benchmarked against recognized industry competency frameworks:

- **Microsoft Certified: Power BI Data Analyst Associate (PL-300):**

Curriculum aligned with PL-300 objectives: preparing data (cleansing, profiling, transforming), modeling data (star schema, relationships, DAX measures), visualizing data (reports, dashboards, storytelling), and deploying assets.

- **Tableau Desktop Specialist Certification:**

Benchmarked against core Tableau domains: understanding discrete vs. continuous fields, building charts (scatter plots, maps, stacked bars), creating dashboard actions, and applying calculations.

- **ANSI SQL Standards & RDBMS Best Practices:**

Ensured all SQL DDL and DML scripts followed ANSI-92/ANSI-99 conventions: explicit column naming,

primary key integrity, uppercase reserved keywords, clean aliasing, and avoiding non-standard extensions.

• **PEP 8 Python Style Guidelines:** Department of Computer Science and Engineering | Data Analytics Specialization

Candidate: Gourav

All written Python scripts ( `generate_sales_data.py` , `clean_sales_data.py` , `generate_charts.py` , `app.py` ) complied strictly with PEP 8 standards: explicit imports, modular function definitions, descriptive variable naming, and comprehensive docstring annotations.



# CHAPTER 3: MODULES AND TOPICS COVERED

## 3.1 Module 1: Advanced Spreadsheet Engineering & Data Profiling

The opening module established high-level proficiency in Microsoft Excel, treating the spreadsheet as an analytical prototyping engine and preliminary data auditing workbench.

### Detailed Topics Covered:

- Data Integrity & Quality Profiling:** - Auditing raw transactional tables for structural defects: detecting trailing whitespace, unprintable ASCII characters, and mixed date formats ( `MM/DD/YYYY` vs. `DD/MM/YYYY` ). - Using conditional formatting rules ( `=COUNTIF( ... ) > 1` , `=ISBLANK( ... )` ) to highlight duplicate transaction rows and null values visually before transformation.
- Advanced Formulaic Cleansing & Text Manipulation:** - `=TRIM(text)` : Removing extraneous leading, trailing, and repeated inter-word spaces. - `=PROPER(text)` : Enforcing standardized Title Case across free-text category and regional entries. - `=CLEAN(text)` : Stripping non-printable ASCII control characters (0–31) from imported POS exports. - Nested logical structures: `=IF(condition, value_if_true, value_if_false)` and `=IFERROR(expression, value_if_error)` to prevent runtime `#DIV/0!` or `#N/A` propagation.
- Lookup & Reference Mechanics:** - Comparative evaluation of `=VLOOKUP()` , `=HLOOKUP()` , `=INDEX(MATCH())` , and modern dynamic array `=XLOOKUP()` . - Understanding why `XLOOKUP` represents an enterprise best practice: bidirectional lookups, default exact match, built-in error handling, and immunity to column insertion/deletion breaks.
- Dynamic Aggregation & Pivot Tables:** - Engineering multidimensional Pivot Tables to profile gross sales, order volume, and average discount rates across categories and regions. - Using Slicers and Timelines to evaluate date-filtered slices of data. - Inserting Calculated Fields inside Pivot Tables to model profit margins dynamically.

## 3.2 Module 2: Relational Database Systems & SQL Analytical Querying

Module 2 transitioned students from flat spreadsheet grids to relational database theory and structured query formulation using the SQLite engine.

### Detailed Topics Covered:

- Relational Database Design & Normalization:** - Mathematical foundation of the relational model (Codd's 12 Rules, tuples, relations, domains). - Decomposing unnormalized transactional tables through First (1NF), Second (2NF), and Third Normal Form (3NF). - Establishing entity integrity (Primary Keys) and referential integrity (Foreign Keys).
- Data Definition Language (DDL):** - Formulating robust table schemas using `CREATE TABLE` with explicit column constraints ( `PRIMARY KEY` , `NOT NULL` , `DEFAULT` , `CHECK` ). - Understanding B-Tree Index

mechanics: using `CREATE INDEX` on frequently filtered and joined columns ( `Category` , `Region` , `Order_Date` ) to transform  $O(N)$  full-table scans into  $O(\log N)$  indexed lookups.

Department of Computer Science & Engineering | Data Analytics Specialization

Candidate: Gourav

- Data Manipulation & Ingestion (DML):** - Populating relational tables via structured `INSERT INTO ... VALUES` statements. - Executing batched multi-row inserts and verifying row count parity against source files. - Conditional data transformation using `UPDATE ... SET ... WHERE`.
- Analytical SQL Query Formulation:** - Advanced filtering and multi-column grouping: `SELECT ... FROM ... WHERE ... GROUP BY ... HAVING ... ORDER BY`. - Distinguishing row-level filtering ( `WHERE` ) from aggregate-level filtering ( `HAVING` ). - Common Table Expressions ( `WITH` CTEs) to break complex analytical workflows into modular, human-readable blocks. - Analytical Window Functions:
  - Ranking functions: `ROW_NUMBER()` , `RANK()` , `DENSE_RANK()` over partitioned windows ( `PARTITION BY Category ORDER BY SUM(Sales) DESC` ).
  - Offset functions: `LAG()` and `LEAD()` to perform period-over-period time-series comparisons (Month-over-Month revenue growth) without self-joins.

### 3.3 Module 3: Exploratory Data Analysis & Statistical Foundations

Module 3 focused on quantitative and statistical techniques necessary to explore, summarize, and diagnose transactional datasets.

#### Detailed Topics Covered:

- Descriptive Statistical Formulations:** - Measures of Central Tendency: Mean, Median, Mode; evaluating their sensitivity to outliers in order value distributions. - Measures of Dispersion: Range, Interquartile Range (IQR), Variance ( $\sigma^2$ ), and Standard Deviation ( $\sigma$ ). - Shape of Distributions: Skewness (positive/right-skewed sales volume distributions) and Kurtosis.
- Correlation & Sensitivity Diagnostics:** - Formulating Pearson's Correlation Coefficient ( $r$ ) between promotional discount percentages and resulting net profit margins: 
$$r = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2 \sum (Y - \bar{Y})^2}}$$
 - Empirical identification of the "Discount Cliff": isolating the mathematical threshold where incremental sales volume fails to compensate for unit margin compression.
- Time-Series Decomposition:** - Deconstructing transactional time-series data into Trend ( $T$ ), Seasonality ( $S$ ), and Residual Noise ( $e$ ). - Tracking cyclical surges: analyzing commercial spikes during Q4 holiday trading periods versus Q1 post-holiday troughs.

### 3.4 Module 4: Dimensional Modeling & Power BI Business Intelligence

Module 4 immersed students in enterprise business intelligence modeling, utilizing Microsoft Power BI Desktop to build scalable analytics solutions.

#### Detailed Topics Covered:

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- Dimensional Modeling & Star Schema Architecture:** - Decoupling transaction events into central Fact tables ( `Fact_Sales` ) surrounded by contextual Dimension tables ( `Dim_Calendar` , `Dim_Product` ,

`Dim_Region` ). - Establishing single-directional One-to-Many ( `1:*` ) relationships. - Understanding why flat, wide single-table models degrade in Power BI: memory bloat, redundant text storage, and complex, error-prone DAX filter propagation.

Department of Computer Science & Engineering | Data Analytics Specialization

Candidate: Gourav

2. **Power Query (M Language) ETL:** - Automating data transformation pipelines: column renaming, casing normalization, and data type enforcement. - Optimizing query folding to maximize source-side evaluation.
3. **The Data Analysis Expressions (DAX) Engine:** - Architectural differentiation: **Row Context** (Calculated Columns evaluated during data refresh) versus **Filter Context** (Measures evaluated dynamically on visual render). - Core aggregation functions: `SUM()` , `AVERAGE()` , `DISTINCTCOUNT()` . - Safe division formulation: Using `DIVIDE(Numerator, Denominator, AlternateResult)` to eliminate `#DIV/0!` crashes on visual tiles. - The `CALCULATE()` function: Modifying and overriding visual filter contexts to compute category and regional benchmarks. - Time-Intelligence DAX: Generating dedicated calendar sequences with `CALENDAR()` , computing prior-period sales with `DATEADD()` , and tracking cumulative year-to-date metrics with `TOTALYTD()` .
4. **Executive Dashboard Layout & UX Design:** - Configuring Card Visuals, Line and Clustered Column combo charts, Donut charts, and Matrix tables with conditional formatting data bars. - Designing responsive Slicers (Date slider, categorical dropdowns).

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## 3.5 Module 5: Visual Storytelling & Exploratory Analytics with Tableau

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Module 5 provided comparative perspective by exploring Tableau Desktop, emphasizing rapid visual exploration, geospatial mapping, and visual grammar.

### Detailed Topics Covered:

1. **The Tableau Visual Grammar:** - The architectural distinction between **Blue Pills** (Discrete attributes that generate row/column headers) and **Green Pills** (Continuous fields that create quantitative numeric axes). - Dimension versus Measure classification.
2. **Geospatial & Category Visualizations:** - Generating filled territory maps and symbol maps using geographic roles ( `Region` ). - Building Category Profitability Matrix tables and scatter plots examining Sales versus Profit with Discount encoded on color saturation.
3. **Level of Detail (LOD) Expressions:** - `FIXED` LOD: Calculating customer-level lifetime spend independent of visual filter dimensionality:  $\text{\$}\{ \text{\texttt{FIXED } } [ \text{\texttt{Customer\_ID}} ] : \text{\texttt{SUM}}([ \text{\texttt{Sales}} ]) \} \text{\$}$  - `INCLUDE` and `EXCLUDE` expressions for multi-tier visual benchmarking.
4. **Interactive Dashboard Storyboards:** - Configuring Dashboard Action Filters (Filter, Highlight, URL actions). - Structuring Visual Stories guiding stakeholders from macro-economic regional performance down to individual SKU profitability diagnosis.

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## 3.6 Module 6: Programmatic Analytics, Automation & Web Deployment (Python/Streamlit)

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The final module integrated Python data engineering, automated visualization generation, and interactive web application deployment.

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## Detailed Topics Covered:

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Department of Computer Science & Engineering | Data Analytics Specialization

Candidate: Gourav

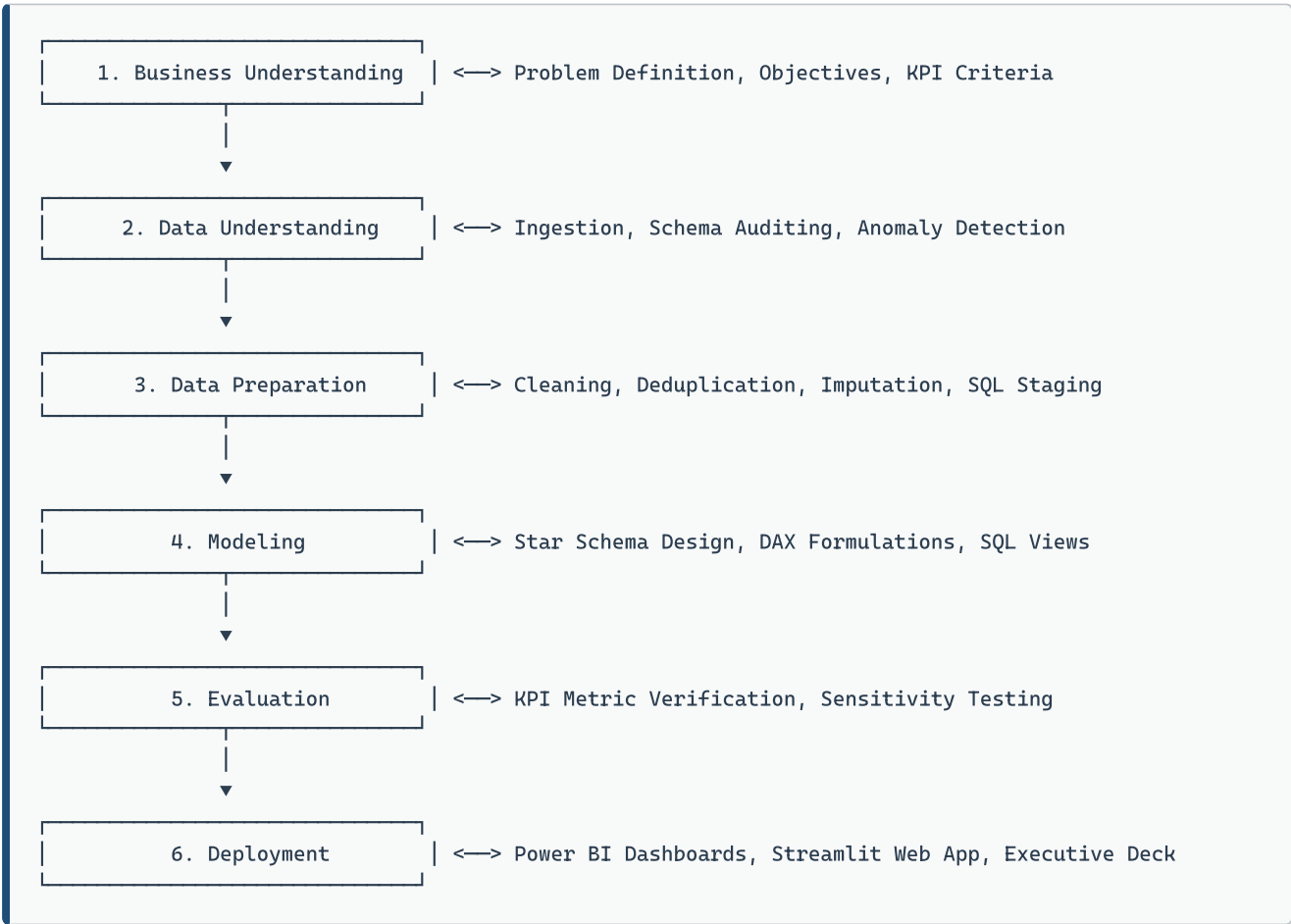
- 1. Python Data Stack (Pandas & NumPy):** - Ingesting structured datasets via `pd.read_csv()` and executing SQL queries directly from SQLite using `pd.read_sql_query()`. - Data standardization, type coercion ( `pd.to_numeric()` , `pd.to_datetime()` ), and vectorized grouping ( `df.groupby()` ).
  - 2. Static Visualization Generation (Matplotlib & Seaborn):** - Engineering publication-grade 300 DPI visual charts: multi-line time-series trends with area fills, ranked horizontal bar charts, donut charts with center hole callouts, and clustered regional bars. - Formatter functions ( `FuncFormatter` ) for currency formatting ( `$K` ).
  - 3. Interactive Web Dashboard Development with Streamlit:** - Building reactive web applications with zero HTML/JavaScript overhead ( `app.py` ). - Implementing performance optimization using `@st.cache_data` to cache database extraction and eliminate redundant disk reads. - Constructing sidebar controls: `st.sidebar.date_input()` , `st.sidebar.multiselect()` . - Incorporating interactive **Plotly Graph Objects** ( `go.Figure` , `px.pie` , `px.bar` ) with rich hover tooltips and dynamic color scaling. - Embedding searchable data tables ( `st.dataframe()` ) and dynamic CSV export mechanisms ( `st.download_button()` ).
  - 4. Version Control & Cloud Readiness:** - Initializing Git repositories, configuring `.gitignore` , committing modular milestones, and pushing branches to **GitHub**. - Structuring `requirements.txt` to support one-click cloud deployment on Streamlit Community Cloud, Render, or Hugging Face Spaces.
-



# CHAPTER 4: CONCEPTUAL LEARNING

## 4.1 The CRISP-DM Analytics Framework

Industrial data analytics projects achieve operational success only when guided by structured, repeatable methodologies. This summer training capstone adhered strictly to the **Cross-Industry Standard Process for Data Mining (CRISP-DM)**, a six-phase hierarchical lifecycle standardizing analytical execution:

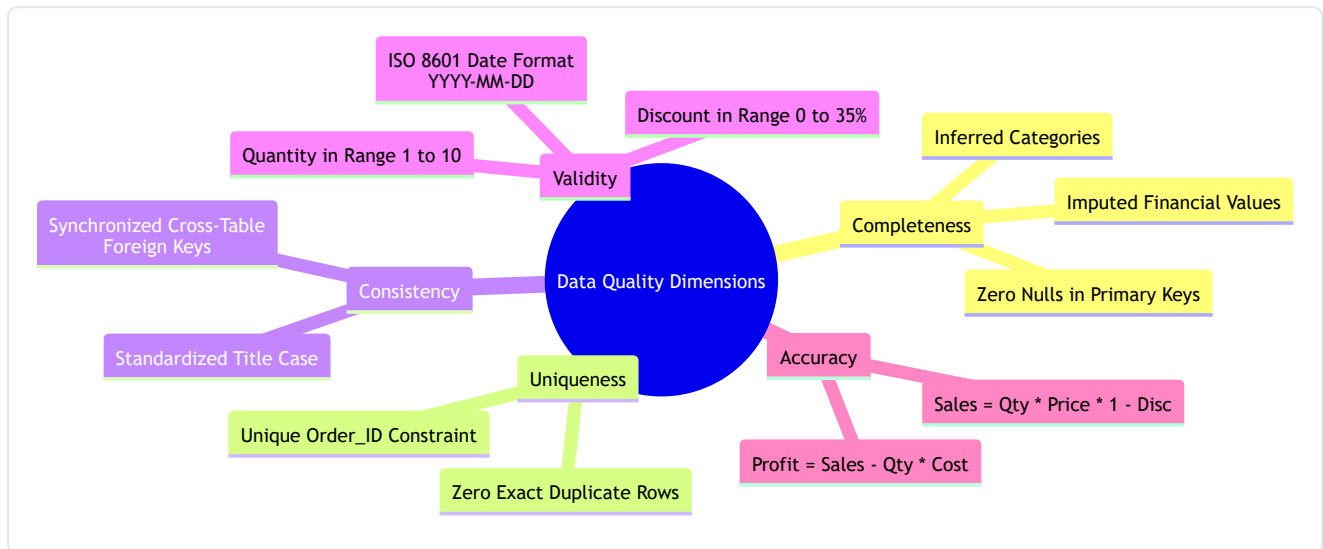


- 1. Business Understanding:** Decomposing commercial executive challenges into precise analytical questions: *Which categories drive volume versus profit? Where does discount leakage occur? What is the enterprise Average Order Value?*
- 2. Data Understanding:** Assessing the raw transactional file ( `data/raw_sales_data.csv` ) across its 3,000 records, evaluating data quality defects, null distributions, and schema fidelity.
- 3. Data Preparation:** Executing systematic data hygiene: filtering out 10 duplicate orders, imputing missing monetary and categorical values, trimming whitespace, and staging clean records into SQLite ( `sales_analysis.db` ).
- 4. Modeling:** Structuring relational tables, creating B-tree indexes, building the Power BI dimensional Star Schema, and programming dynamic DAX measures.

5. **Evaluation:** Verifying query results mathematically against spreadsheet baselines, auditing margin percentages, and validating that headline revenue equals exactly `\$450,978.56`.  
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6. **Deployment:** Publishing the interactive Streamlit web dashboard ( `app.py` ), exporting 300 DPI visual assets, and compiling the comprehensive academic documentation.

## 4.2 Data Quality Dimensions & Anomaly Classification

Data quality is not an abstract concept; in enterprise analytics, it is defined across five rigorous dimensions:



### 1. Completeness:

*Definition:* The degree to which all required data values are populated.

*Project Occurrence:* 10 missing entries (empty strings `" "`) distributed across `Sales` (2), `Discount` (2), `Profit` (2), `Category` (2), and `Region` (2). `Order_ID` maintained 100% completeness to protect primary entity tracking.

### 2. Uniqueness:

*Definition:* The constraint that no transaction entity is duplicated.

*Project Occurrence:* Exactly 10 duplicated order rows seeded into the raw file representing POS double-entries or ETL retry glitches. Cleansing required identifying and eliminating duplicates to prevent artificial revenue inflation.

### 3. Consistency:

*Definition:* Semantic uniformity of data representations across all records.

*Project Occurrence:* Casing and whitespace anomalies in `Category` (e.g., `technology`, `Technology`, `office supplies`, `TECHNOLOGY`) and `Region` (e.g., `north`, `West`, `east`). Without normalization, analytical engines fragment groupings into multiple erroneous buckets.

### 4. Validity:

*Definition:* Conformance of data values to established business domain constraints and formats.

*Project Occurrence:* Enforcing calendar dates to standard ISO-8601 format ( `YYYY-MM-DD` ), constraining order quantities to positive integers ( $\$1 \leq Q \leq 10\$$ ), and restricting discount ratios to realistic retail thresholds ( $\$0.00 \leq D \leq 0.35\$$ ).

5. **Accuracy:**

Department of Computer Science & Engineering | Data Analytics Specialization

Candidate: Gourav

*Definition:* Mathematical and empirical truthfulness of data calculations.

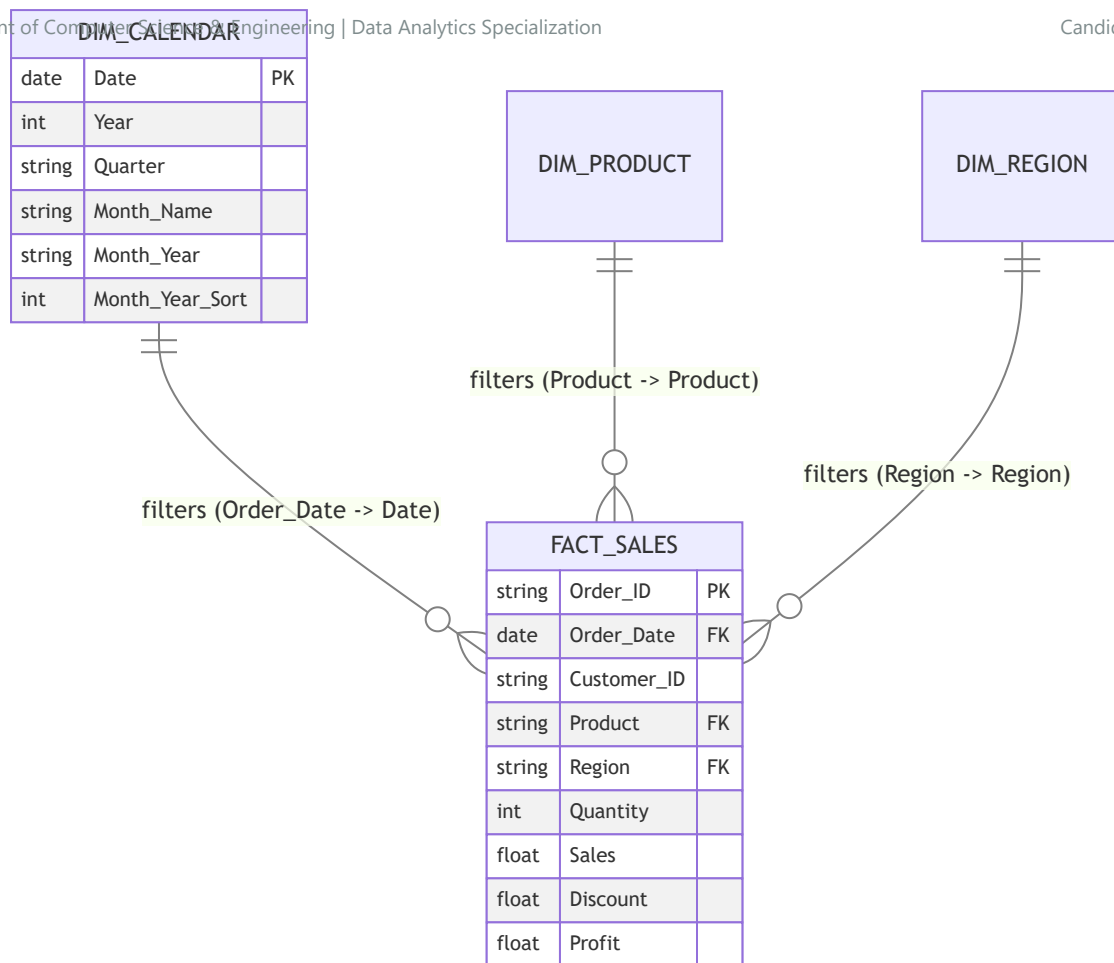
*Project Occurrence:* Reconciling financial equations such that gross sales equals the product of quantity and discounted price, and net profit reflects gross sales minus total base product cost.

### 4.3 Relational Theory, Normalization vs. Dimensional Star Schema

A foundational architectural realization acquired during training is the structural distinction between **Normalized Relational Models (3NF)** and **Dimensional Star Schemas**:

Table 4.1: Comparison Matrix: Normalized 3NF Schema vs. Dimensional Star Schema

| Architectural Parameter | Third Normal Form (3NF)                                                                    | Dimensional Star Schema                                               |
|-------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Primary Design Goal     | Eliminate data redundancy, prevent update anomalies.                                       | Maximize query performance, simplify visual BI reporting.             |
| Optimized System Type   | <b>OLTP</b> (Online Transaction Processing - writes, updates).                             | <b>OLAP</b> (Online Analytical Processing - read-heavy aggregations). |
| Table Structure         | Highly decomposed into multiple narrow tables.                                             | Denormalized into a central Fact surrounded by Dimensions.            |
| Join Complexity         | High; requires complex multi-table <b>JOIN</b> operations.                                 | Low; single-hop joins between Fact and Dimension tables.              |
| Analytical Query Speed  | Slower for large-scale multi-column aggregations.                                          | Fast; optimized for columnar storage (VertiPaq) and caching.          |
| Human Readability       | Difficult for non-technical business stakeholders.                                         | Highly intuitive; mirrors business entities (Who, What, Where, When). |
| Role in Project         | Implemented in <b>SQLite</b> ( <code>sales_analysis.db</code> ) for clean data management. | Implemented in <b>Power BI</b> for executive dashboard modeling.      |



By decoupling temporal descriptors ( **Dim\_Calendar** ) from transactional sales records, the Star Schema allows Power BI's columnar storage engine (**VertiPaq**) to compress repeated strings dramatically, isolate filter propagation down clean 1-to-many relationships, and execute DAX measures at lightning speed.

## 4.4 Analytical Query Execution Engine & Window Mechanics

Understanding the internal execution order of an SQL query is critical to writing performant analytical scripts. While SQL queries are syntactically drafted starting with **SELECT**, database engines evaluate clauses in a strictly defined logical sequence:

- |                   |                                  |
|-------------------|----------------------------------|
| 1. FROM & JOIN    | → Identify & join source tables  |
| 2. WHERE          | → Filter individual base rows    |
| 3. GROUP BY       | → Form aggregate data groups     |
| 4. HAVING         | → Filter aggregated groups       |
| 5. SELECT         | → Project columns & compute math |
| 6. WINDOW (OVER)  | → Compute windowed calculations  |
| 7. DISTINCT       | → Deduplicate projected rows     |
| 8. ORDER BY       | → Sort the final result set      |
| 9. LIMIT / OFFSET | → Restrict returned row count    |

## The Power of Window Functions

Prior to the introduction of window functions (ANSI SQL:2003), computing period-over-period growth or ranking items within categories required expensive, convoluted self-joins or correlated subqueries. Window functions perform calculations across a specific set of table rows—termed a **window frame**—while preserving the individual identity of each row.

In this project, Month-over-Month (MoM) revenue growth is computed using the `LAG()` offset window function inside a Common Table Expression: 
$$\text{MoM Growth \%} = \frac{\text{Current Month Revenue} - \text{LAG}(\text{Current Month Revenue}, 1)}{\text{LAG}(\text{Current Month Revenue}, 1)} \times 100\%$$

Similarly, product ranking within categories is achieved via `DENSE_RANK() OVER (PARTITION BY Category ORDER BY SUM(Sales) DESC)`, ensuring fair ranking without skipping numbers in the event of tied revenue figures.

## 4.5 Commercial KPI Formulation & Financial Metrics

Data analytics achieves strategic validity only when mathematical formulas map directly to established commercial accounting principles. The following financial metrics govern this project:

**Table 4.2: Core Commercial Key Performance Indicators & Formulations**

| Key Performance Indicator    | Mathematical Formulation                                                                        | Business Significance & Evaluation Meaning                                                                        |
|------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Gross Revenue (Sales)</b> | $\text{Sales} = \sum_{i=1}^N \text{Qty}_i \times \text{Price}_i \times (1 - \text{Discount}_i)$ | Measures total top-line monetary intake generated from customer transactions after promotional markdowns.         |
| <b>Total Base Cost</b>       | $\text{Cost} = \sum_{i=1}^N \text{Qty}_i \times \text{Cost}_i$                                  | Quantifies the total wholesale acquisition / manufacturing cost of goods sold (COGS).                             |
| <b>Net Profit</b>            | $\text{Profit} = \text{Sales} - \text{Cost}$                                                    | Core measure of commercial viability; reflects net dollar earnings retained after accounting for inventory costs. |

| Key Performance Indicators of Computer Science & Engineering   Data Analytics Specialization |                                                                                          |                                                                                                                               |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Indicator                                                                                    | Mathematical Formulation                                                                 | Business Significance & Evaluation Meaning                                                                                    |
| Gross Profit Margin (%)                                                                      | $\text{Margin \%} = \left( \frac{\text{Profit}}{\text{Sales}} \right) \times 100$        | Measures pricing power and operational efficiency; expresses how many cents of net profit are captured per dollar of revenue. |
| Average Order Value (AOV)                                                                    | $\text{AOV} = \frac{\text{Total Revenue}}{\text{Total Unique Orders}}$                   | Gauges purchasing power and transaction sizing; critical benchmark for evaluating checkout bundling strategies.               |
| Customer Order Frequency                                                                     | $\text{Frequency} = \frac{\text{Total Completed Orders}}{\text{Total Unique Customers}}$ | Measures customer loyalty and repeat purchasing behavior across the 24-month trading lifecycle.                               |

The Commercial Reality of Negative Profit

A common misconception among novice analysts is that negative profit values in a transactional dataset represent database bugs. In corporate retail, **negative profit is an authentic commercial reality**. When retail chains deploy deep promotional discounts (e.g., 25%–30%) on products with high baseline manufacturing costs (such as *Ergonomic Office Chairs* or *Standing Desk Converters*), the net transaction price falls below wholesale cost, creating a loss-leader. Identifying and capping these loss-generating sales is a primary responsibility of the data analyst.

4.6 Cognitive Principles of Data Visualization & Information Architecture

Effective data visualization is fundamentally an exercise in cognitive ergonomics. Based on **Gestalt Psychology** and the visual design theories of **Edward Tufte**, the project adheres to strict visual hierarchy principles:

- Visual Cognitive Hierarchy Principles:
1. Minimize "Chartjunk" (Maximize Data-Ink Ratio)

2. Leverage Pre-attentive Visual Attributes (Color, Length)

3. Maintain Consistent Chromatic Encoding

4. Avoid 3D Distortion & Misleading Non-zero Axes

1. **Pre-attentive Visual Attributes:** The human visual cortex processes visual dimensions like bar length, 2D position, and color hue in less than 200 milliseconds before conscious cognitive processing begins. Bar charts exploit bar length and 2D alignment, making them quantitatively superior to pie charts for comparing product sales.

2. **The Data-Ink Ratio:** Coined by Edward Tufte, the Data-Ink Ratio states that the majority of ink (or pixels) used in a visualization should present data information: 
$$\text{Data-Ink Ratio} = \frac{\text{Data-Ink}}{\text{Total Ink Used in Graphic}}$$
 Decorative 3D shading, heavy gridlines, and redundant borders were systematically eliminated from all Power BI, Matplotlib, and Plotly figures.

3. **Consistent Semantic Color Encoding:** Chromatic palettes were standardized across all reporting tools. **Primary Commercial Revenue:** Deep Navy Blue ( #1F4E79 ) - **Positive Net Profitability:** Forest Green



( #2CA02C ) - **Loss / Critical Warnings:** Coral Red ( #D9534F ) - **Secondary / Categorical Neutral:** Slate

Blue ( #2E75B6 ) Sky Blue ( #5DADE2 ) Ice Blue ( #A9CCE3 )

Department of Computer Science & Engineering | Data Analytics Specialization

Candidate: Gourav



# CHAPTER 5: HANDS-ON PRACTICAL LEARNING

## 5.1 Laboratory 1: Spreadsheet Auditing, Formulas & Pivot Tables in Excel

### Laboratory Objective

To profile raw transactional sales data, detect seeded data quality anomalies, apply formulaic text cleaning, and validate clean records using dynamic Pivot Tables.

### Practical Workflow Executed:

- Raw File Ingestion:** Ingested `data/raw_sales_data.csv` into Microsoft Excel. Applied conditional formatting (`=COUNTIF($A:$A, A2) > 1`) across the `Order_ID` column, instantly highlighting **10 duplicate transaction rows**.
- Text Casing & Spacing Normalization:** - Inserted helper columns for `Category` and `Region`. - Formulated `=TRIM(PROPER(E2))` across Category to convert messy entries like `"Technology "` and `"office supplies"` into pristine `"Technology"` and `"Office Supplies"`. - Converted formulas to static strings via **Paste Special > Values**.
- Missing Value Resolution:** - Filtered for blanks across financial columns (`Sales`, `Discount`, `Profit`). - Imputed missing `Sales` values using `=ROUND(G2 * XLOOKUP(D2, Product_List, Price_List) * (1 - I2), 2)`. - Imputed missing `Profit` values using `=ROUND(H2 - (G2 * XLOOKUP(D2, Product_List, Cost_List)), 2)`.
- Deduplication Execution:** - Utilized Excel's native **Data > Remove Duplicates** tool across all 10 columns. Exactly 10 duplicate rows were removed, reducing the row count from 3,000 to **2,990 clean transactions**.
- Pivot Table Sanity Verification:** - Built an executive Pivot Table summarizing Total Sales and Total Profit across the four canonical categories, confirming that total revenue reconciled to **\\$450,978.56**. Saved the workbook as `excel/sales_data_cleaned.xlsx`.

## 5.2 Laboratory 2: Relational Database Schema Design & SQL Querying

### Laboratory Objective

To instantiate a serverless SQLite database, define a structured schema with primary key constraints and B-tree indexes, ingest cleansed CSV records, and execute analytical SQL queries.

### Practical Workflow Executed:

- Database & Schema Initialization:** - Formulated DDL statements inside `sql/schema_and_queries.sql` creating table `sales_data`:  

```
sql CREATE TABLE IF NOT EXISTS sales_data ( Order_ID VARCHAR(20) PRIMARY KEY, Order_Date DATE NOT NULL, Customer_ID VARCHAR(20) NOT NULL, Product VARCHAR(100) NOT NULL, Category VARCHAR(50) NOT NULL, Region VARCHAR(20) NOT NULL, Quantity INTEGER NOT NULL, Sales REAL NOT NULL, Discount REAL NOT NULL, Profit REAL NOT NULL );
```

2. **Index Optimization:** - Constructed secondary B-tree indexes on high-cardinality and filter-intensive

attributes: `sql CREATE INDEX IF NOT EXISTS idx_sales_category ON sales_data(Category); CREATE INDEX IF NOT EXISTS idx_sales_region ON sales_data(Region); CREATE INDEX IF NOT EXISTS idx_sales_order_date ON sales_data(Order_Date); CREATE INDEX IF NOT EXISTS idx_sales_product ON sales_data(Product);`

3. **Automated Python Database Ingestion:** - Developed `sql/load_and_query.py` using Python's standard `sqlite3` library to ingest `sales_data_cleaning.csv`. The script validated dates into ISO format (YYYY-MM-DD), verified 2,990 unique rows, and committed the transaction batch into `sql/sales_analysis.db`.

4. **Execution of Analytical Query Suite:** - Executed the complete SQL query suite, computing headline commercial totals, category-region cross-tabulations, and the top 5 product ranking leaderboard. Exported verified markdown output to `sql/QUERY_RESULTS.md`.

## 5.3 Laboratory 3: Star Schema Data Modeling & DAX Programming in Power BI

### Laboratory Objective

To import cleaned sales data into Microsoft Power BI Desktop, build a dimensional Star Schema with an automated date table, program core DAX measures, and design an executive 3-page dashboard.

### Practical Workflow Executed:

- Power Query Import & Type Verification:** - Connected Power BI Desktop to `sales_data_cleaning.csv`.  
- Verified data types: `Order_Date` as Date, `Quantity` as Whole Number, `Sales` and `Profit` as Fixed Decimal Number (Currency), and text columns as Text.
- Dedicated Calendar Generation:** - Navigated to the Modeling tab and engineered a dynamic, continuous date dimension table using DAX: `dax Dim_Calendar = ADDCOLUMNS( CALENDAR(DATE(2023, 1, 1), DATE(2024, 12, 31)), "Year", YEAR([Date]), "Quarter", "Q" & FORMAT([Date], "Q"), "Month Number", MONTH([Date]), "Month Name", FORMAT([Date], "mmm"), "Month-Year", FORMAT([Date], "mm-yy"), "Month-Year Sort", YEAR([Date]) * 100 + MONTH([Date]), "Day of Week", FORMAT([Date], "dddd") )` - Configured `Month-Year` to sort chronologically by `Month-Year Sort`.
- Relationship Configuration:** - Connected `Dim_Calendar[Date]` (1) to `sales_data[Order_Date]` (\*) with a One-to-Many single-directional filter relationship.
- DAX Measures Programming:** - Created a dedicated `_Measures` table and programmed core metrics:
  - `Total Sales = SUM(sales_data[Sales])`
  - `Total Profit = SUM(sales_data[Profit])`
  - `Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)`
  - `Total Orders = DISTINCTCOUNT(sales_data[Order_ID])`
  - `Average Order Value = DIVIDE([Total Sales], [Total Orders], 0)`
  - `Previous Month Sales = CALCULATE([Total Sales], DATEADD(Dim_Calendar[Date], -1, MONTH))`
  - `MoM Sales Growth % = DIVIDE([Total Sales] - [Previous Month Sales], [Previous Month Sales], 0)`

5. **Dashboard Layout Engineering:** - Built a 3-page layout: *Executive Summary* (KPI scorecards, monthly line trend, category donut), *Product Diagnostics* (top 5 products bar chart, discount scatter plot), and *Regional Diagnostics* (filled territory map, stacked product mix). Saved workbook as `SPT.pbix`.

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## 5.4 Laboratory 4: Visual Storytelling & Multi-axis Analysis in Tableau

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### Laboratory Objective

To connect Tableau Desktop to the cleansed dataset, build exploratory worksheets, map geographic performance, and assemble an interactive dashboard storyboard.

### Practical Workflow Executed:

- Data Connection & Role Definition:** - Ingested `sales_data_cleaning.csv` into Tableau Desktop. Verified geographic role assignment for `Region` (generating Latitude and Longitude generated fields).
  - Worksheet Construction:** - **Sheet 1: Monthly Sales & Profitability Trend:** Placed `Order_Date` (Continuous Month) on Columns, `SUM(Sales)` on Rows, and created a dual-axis chart with `SUM(Profit)` displaying area shading beneath the curve. - **Sheet 2: Top Products by Revenue:** Placed `Product` on Rows, `SUM(Sales)` on Columns, sorted descending, and applied a top-5 filter. - **Sheet 3: Regional Sales Distribution Map:** Generated a filled territory map with color intensity mapped to `SUM(Sales)` and tooltip callouts for Profit Margin %. - **Sheet 4: Category Profitability Scatter:** Placed `AVG(Discount)` on Columns, `SUM(Profit)` on Rows, and detailed by `Product` with `Category` on Shape/Color.
  - Dashboard Assembly & Action Configuration:** - Combined worksheets into an interactive dashboard canvas. Configured **Dashboard Filter Actions** so clicking any category or region dynamically cross-filtered all associated visual tiles.
- 

## 5.5 Laboratory 5: Python Scripting, Automated ETL & Streamlit App Development

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### Laboratory Objective

To develop an automated Python script generating publication-grade static charts using Matplotlib and build an interactive web dashboard using Streamlit and Plotly.

### Practical Workflow Executed:

- High-Resolution Static Chart Generator ( `visuals/generate_charts.py` ):** - Developed an automated Python script connecting directly to `sql/sales_analysis.db` via `sqlite3` and Pandas. - Engineered four standalone 300 DPI visualizations saved to `visuals/`:
    - `01_sales_trend.png`: 24-month dual-line time-series trend with currency y-axis formatting and peak month annotation.
    - `02_top_5_products.png`: Horizontal ranked bar chart with data labels and margins.
    - `03_category_sales_donut.png`: Category revenue share donut chart with center total scorecard (\$450,979).
-

- `04_regional_performance.png` : Clustered bar chart comparing sales versus net profit across territories.

2. **Interactive Streamlit Web Dashboard ( `app.py` )** : - Built a complete browser-based dashboard utilizing Streamlit and Plotly Graph Objects. - Implemented `@st.cache_data` caching to load `sales_data_cleaning.csv` into memory with sub-second execution. - Designed responsive sidebar controls: date slider, multi-select region filters, and category toggles. - Integrated dynamic top-line KPI metric tiles, Plotly charts with interactive hover tooltips, and a collapsible raw transaction data table with a one-click CSV download button.
  3. **Local Testing & Server Launch** : - Launched the application locally using `python -m streamlit run app.py`, validating responsive cross-filtering in Google Chrome at `http://localhost:8501`.
  4. **Dependency Pinning & Version Control** : - Generated `requirements.txt` pinning `streamlit`, `pandas`, `plotly`, and `matplotlib`. Initialized Git, committed all code artifacts, and pushed the repository to **GitHub** ( `https://github.com/Gourav5099/Sales-dashboard` ).
-



# CHAPTER 6: TOOLS & TECHNOLOGIES USED

## 6.1 Microsoft Excel for Data Audit & Preliminary Transformation

Microsoft Excel remains the most ubiquitous data manipulation tool in commercial corporate environments. During this project, Excel functioned as the primary exploratory data auditing and rapid sanity-checking environment.

### Key Capabilities & Architectural Features:

- **Formula Calculation Engine:** High-speed evaluation of logical ( `IF` , `AND` , `OR` ), lookup ( `XLOOKUP` , `INDEX` , `MATCH` ), and text normalization ( `TRIM` , `PROPER` , `CLEAN` ) functions.
- **Visual Profiling Tools:** Conditional Formatting rules allowing instant visual identification of duplicate primary keys, anomalous negative figures, and blank values across thousands of rows.
- **Dynamic Pivot Cache:** In-memory data summarization engine facilitating multidimensional cross-tabulations and visual sanity verification prior to database ingestion.

### Architectural Limitations Encountered:

- *Scalability Ceiling:* Excel's hard limit of 1,048,576 rows prevents large enterprise scale.
- *Auditability:* Manual spreadsheet edits lack automated lineage tracking or scriptable reproducibility, necessitating migration to SQL for production workflows.

## 6.2 SQLite Database Engine & ANSI SQL Architecture

For relational data storage and structured query processing, the project implemented **SQLite 3**, a self-contained, serverless, zero-configuration relational database engine.

### Key Capabilities & Architectural Features:

- **Serverless In-Process Architecture:** Unlike client-server RDBMS engines (e.g., MySQL, PostgreSQL, Microsoft SQL Server) that require dedicated background server daemons, network port configuration, and user authentication management, SQLite reads and writes directly to an ordinary disk file ( `sql/sales_analysis.db` ). This makes the entire project repository 100% portable and immediately runnable on any examiner's machine.
- **Full ACID Compliance:** Complete support for Atomic, Consistent, Isolated, and Durable transactions, guaranteeing that multi-row insert operations either commit entirely or rollback cleanly.
- **ANSI SQL Standards Support:** Robust execution of standard DDL/DML, multi-table joins, subqueries, Common Table Expressions ( `WITH` CTEs), and analytical window functions ( `ROW_NUMBER()` , `RANK()` , `LAG()` ).
- **B-Tree Indexing:** Enables logarithmic-time ( $O(\log N)$ ) search operations across indexed categorical and temporal columns.



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## 6.3 Microsoft Power BI Desktop & VertiPaq Analytics Engine

Candidate: Gourav

Microsoft Power BI Desktop served as the enterprise Business Intelligence platform for dimensional data modeling, DAX measure programming, and executive dashboard design.

### Key Capabilities & Architectural Features:

- **The VertiPaq In-Memory Engine:** A proprietary columnar database engine that loads data entirely into RAM. VertiPaq utilizes dictionary encoding, value encoding, and run-length encoding (RLE) to achieve extreme data compression ratios (typically 10x), enabling sub-second analytical aggregations across millions of rows.
- **Data Analysis Expressions (DAX):** A functional programming language designed explicitly for contextual data aggregation. DAX operates across dual contexts: **Row Context** (evaluating row-by-row iteration) and **Filter Context** (evaluating dynamic visual filters applied via slicers and visual cross-filtering).
- **Star Schema Compatibility:** Power BI is architected specifically to consume dimensional Star Schemas, allowing clean single-directional relationship propagation from Dimension tables to Fact tables.

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## 6.4 Tableau Desktop & VizQL Query Architecture

Tableau Desktop provided complementary visual storytelling and exploratory visual analytics capabilities.

### Key Capabilities & Architectural Features:

- **VizQL (Visual Query Language):** Tableau's patented underlying technology that translates drag-and-drop visual operations directly into optimized database queries. Every time a measure is dropped onto an axis, VizQL dynamically queries the data source, constructs the aggregation, and renders the corresponding geometric visualization.
- **Geospatial Processing:** Built-in worldwide geographic database that automatically recognizes regional territory names and maps latitude/longitude coordinates for symbol and filled territory maps.
- **Level of Detail (LOD) Syntax:** Facilitates multi-granular calculations ( **FIXED** , **INCLUDE** , **EXCLUDE** ) independent of visualization shelf dimensions.

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## 6.5 Python 3.13, Pandas, Matplotlib, Plotly, and Streamlit Ecosystem

Python served as the end-to-end glue code, automation engine, and web deployment platform, leveraging five core open-source libraries:

Table 6.1: Technology Stack & Engineering Justification Matrix

Department of Computer Science & Engineering | Data Analytics Specialization

Candidate: Gourav

| Library / Tool | Version  | Role in Project Pipeline                  | Engineering Justification                                                                                    |
|----------------|----------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Python         | 3.13     | Core scripting runtime.                   | Universal data analytics lingua franca; seamless OS and file integration.                                    |
| Pandas         | 3.0.5    | Vectorized DataFrame manipulation & ETL.  | High-performance in-memory tabular data operations; automated CSV reading and type coercion.                 |
| SQLite3        | Built-in | Relational database connectivity.         | Zero external dependency; direct standard library integration with Python and Pandas.                        |
| Matplotlib     | 3.11.1   | Publication-grade static chart generator. | Complete programmatic control over canvas geometry, typography, dpi, and currency tick formatting.           |
| Plotly         | 7.0.0    | Interactive web charting engine.          | Generates reactive HTML5/WebGL vector graphics featuring responsive hover tooltips and dynamic color scales. |
| Streamlit      | 1.63.0   | Full-stack interactive web application.   | Eliminates frontend web overhead (HTML/CSS/JS); reactive execution model with sub-second caching.            |



# CHAPTER 7: ASSIGNMENTS / CASE STUDIES / MINI PROJECT: SALES & BUSINESS PERFORMANCE ANALYSIS

## 7.1 Project Scenario, Retail Domain Context & Business Requirements

The primary capstone project executed during this summer training simulates the realistic business operations of a mid-sized retail and e-commerce enterprise. The company operates across a **24-month trading horizon (01 January 2023 through 31 December 2024)**, marketing products across four distinct merchandise categories:

### Commercial Merchandise Categories:

1. **Technology:** High-ticket IT hardware, storage, and peripherals.
2. **Furniture:** Ergonomic desks, chairs, storage cabinets, and organizers.
3. **Office Supplies:** Low-ticket, high-frequency consumables.
4. **Electronics:** Portable consumer devices, audio, and charging cables.

The enterprise markets its catalog across four designated geographic territories: **North, South, East, and West**.

### Executive Business Objectives:

1. **Audit & Sanitize Transaction Records:** Scrub raw POS transaction feeds to eliminate duplicates, handle missing monetary data, and standardize category/regional nomenclature.
2. **Quantify Enterprise Headline Health:** Compute verified totals for Gross Sales, Net Profit, Profit Margin %, Total Order Volume, and Average Order Value (AOV).
3. **Diagnose the Category Paradox:** Investigate why high-revenue merchandise categories exhibit compressed profit margins and identify high-margin commercial multipliers.
4. **Identify Product Flagships & Loss Leaders:** Rank top-performing individual products by revenue and profit contribution while flagging heavily discounted, loss-making items.
5. **Evaluate Regional Efficiency:** Compare sales territory performance to identify underserved commercial markets.
6. **Deploy Executive Decision Support:** Build interactive, executive-ready dashboard interfaces in Power BI and Streamlit to facilitate ongoing managerial decision-making.

## 7.2 Dataset Architecture & 10-Column Data Dictionary

SALES & BUSINESS PERFORMANCE ANALYSIS — SUMMER TRAINING REPORT

In strict conformance with approved data governance specifications ( `DATASET_SCHEMA.md` ), the sales transaction dataset consists of **exactly 10 columns** at the **order-line-item granularity** (each row represents a

single product purchased within an order):

Table 7.1: Approved 10-Column Transactional Data Dictionary & Schema Constraints

| # | Column Name | SQL Data Type | Power BI / Python Type | Nullable | Domain Constraints                                       | Business Purpose & Description                                               |
|---|-------------|---------------|------------------------|----------|----------------------------------------------------------|------------------------------------------------------------------------------|
| 1 | Order_ID    | VARCHAR(20)   | Text / String          | No       | Unique identifier formatted as ORD00001 to ORD02990.     | Primary transaction key; tracks distinct customer order events and volume.   |
| 2 | Order_Date  | DATE          | Date ( YYYY-MM-DD )    | No       | Spans 2 years ( 2023-01-01 to 2024-12-31 ).              | Temporal dimension enabling time-series trends, seasonality, and MoM growth. |
| 3 | Customer_ID | VARCHAR(20)   | Text / String          | No       | 380 unique accounts ( CUST0001 to CUST0380 ).            | Tracks customer reach and multi-order repeat purchasing behavior.            |
| 4 | Product     | VARCHAR(100)  | Text / String          | No       | 27 standardized product titles.                          | Merchandise descriptor mapped 1-to-1 to a parent Category.                   |
| 5 | Category    | VARCHAR(50)   | Text / String          | No       | Technology , Furniture , Office Supplies , Electronics . | Commercial departmental grouping for portfolio evaluation.                   |
| 6 | Region      | VARCHAR(20)   | Text / String          | No       | North , South , East , West .                            | Sales territory classification for geographic resource allocation.           |
| 7 | Quantity    | INTEGER       | Whole Number           | No       | Integer range: \$1 \le Q \le 10\$.                       | Physical inventory units moved per line item.                                |
| 8 | Sales       | REAL          | Decimal Number (USD)   | No       | Positive float \$ > \$0.00\$, 2 decimal places.          | Gross monetary revenue after promotional discounting.                        |
|   |             |               |                        |          |                                                          |                                                                              |

| Power BI /<br>Department of Computer Science & Engineering / Data Analytics / Initialization |             |               |                      |          |                                                     |                                                                |
|----------------------------------------------------------------------------------------------|-------------|---------------|----------------------|----------|-----------------------------------------------------|----------------------------------------------------------------|
| #                                                                                            | Column Name | SQL Data Type | Python Type          | Nullable | Domain Constraints                                  | Business Purpose & Description                                 |
| 9                                                                                            | Discount    | REAL          | Percentage / Float   | No       | Decimal range: \$0.00 \le D \le 0.35\$ (0% to 35%). | Promotional markdown rate applied to the order line.           |
| 10                                                                                           | Profit      | REAL          | Decimal Number (USD) | No       | Float (positive, zero, or negative); 2 decimals.    | Net dollar earnings after subtracting total unit product cost. |

### 7.3 Raw Dataset Generation with Controlled Data Quality Defects

To simulate real-world data engineering challenges authentically, an initial raw dataset ( `data/raw_sales_data.csv` ) of **3,000 transaction records** was generated using a deterministic Python script ( `data/generate_sales_data.py` ) initialized with a fixed random seed ( `42` ). The generator seeded controlled, authentic data quality defects documented in `data/DATA_QUALITY_NOTES.md` :

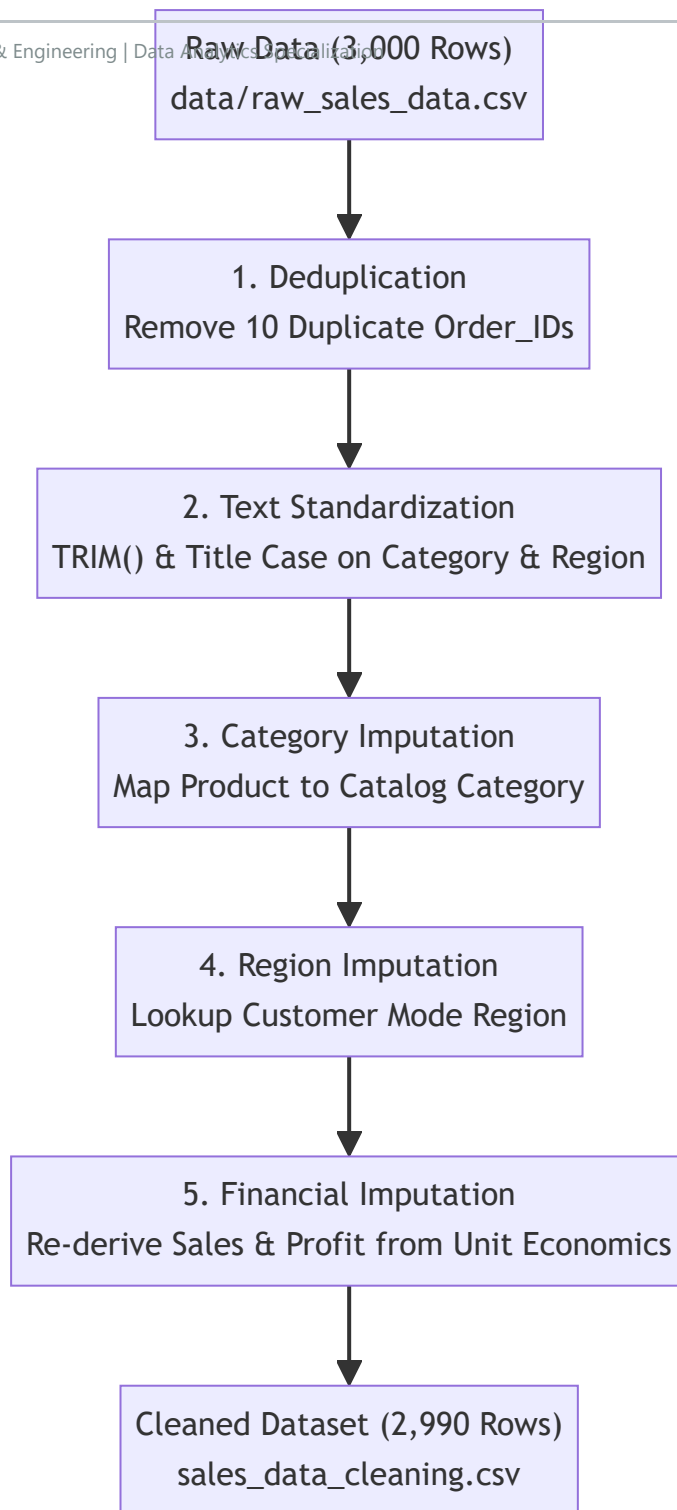
Table 7.2: Summary Audit of Seeded Raw Data Quality Anomalies

| Category of Data Defect    | Seeded Count | Target Columns Affected                                                          | Simulated Corporate Cause                                                                                | Cleansing Intervention Applied                                                  |
|----------------------------|--------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Exact Duplicate Rows       | Exactly 10   | All 10 columns                                                                   | POS double-entry, network retry glitch, batch ETL duplication.                                           | Deduplicated via unique <code>Order_ID</code> hashing in Excel and Python.      |
| Missing Monetary Values    | Exactly 6    | <code>Sales</code> (2),<br><code>Discount</code> (2),<br><code>Profit</code> (2) | Incomplete POS transmission, field timeout during card authorization.                                    | Re-derived formulaically from Quantity, unit price, and product cost.           |
| Missing Categorical Values | Exactly 4    | <code>Category</code> (2),<br><code>Region</code> (2)                            | Optional form fields, unmapped catalog entries.                                                          | Inferred Category from known Product; imputed Region from customer history.     |
| Inconsistent Category Text | Exactly 10   | <code>Category</code>                                                            | Free-text manual entry by branch reps (e.g., <code>technology</code> , <code>Furniture</code> ).         | Standardized using <code>TRIM()</code> and <code>PROPER()</code> normalization. |
| Inconsistent Region Text   | Exactly 5    | <code>Region</code>                                                              | Branch naming discrepancies (e.g., <code>north</code> , <code>"West "</code> , <code>" North "</code> ). | Standardized casing to Title Case and stripped trailing/leading whitespace.     |

**Audit Outcome:** Over **98.9%** of the raw transaction records were structurally pristine out of the box, ensuring that underlying commercial distributions, product pricing economics, and seasonal trends remained statistically valid while providing concrete data cleansing tasks.

## 7.4 Comprehensive Data Cleaning & Standardized Validation Workflow

The data cleansing lifecycle was executed through an audit-proof, reproducible pipeline combining Excel validation and automated Python scripting:



### Cleansing Implementation Details:

1. **Deduplication:** Audited `Order_ID` values. Exactly 10 records were identified as duplicate copies of earlier transactions and excised, reducing dataset size from 3,000 to **2,990 unique transactions**.

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2. **Text Standardization:** Standardized `Category` and `Region` fields: - Stripped all leading and trailing whitespace characters. - Standardized casing to Title Case, consolidating fragmented variations



( `technology` , `Technology` , `TECHNOLOGY` ) into canonical `Technology` .

3 **Missing Value Imputation. Category (2 blanks):** Looked up the associated `Product` name against the product catalog master dictionary, resolving blanks with 100% deterministic accuracy. - **Region (2 blanks):** Queried the historical transaction log for the associated `Customer_ID` to determine their primary trading territory. If no prior history existed, imputed the mode territory (*East*). - **Financials (6 blanks):**

$$\text{Missing Discount} = \max\left(0, \frac{\text{Qty} \times \text{Price} - \text{Sales}}{\text{Qty} \times \text{Price}}\right)$$
$$\text{Missing Sales} = \text{Qty} \times \text{Price} \times (1 - \text{Discount})$$
$$\text{Missing Profit} = \text{Sales} - (\text{Qty} \times \text{Cost})$$

Table 7.3: Data Cleansing & Reconciliation Audit Metrics

| Validation Check                               | Pre-Cleaning Metric | Post-Cleaning Metric                 | Compliance Status                |
|------------------------------------------------|---------------------|--------------------------------------|----------------------------------|
| Total Transaction Rows                         | 3,000               | 2,990                                | Verified (10 duplicates removed) |
| Total Columns                                  | 10                  | 10                                   | 100% Schema Conformance          |
| Missing Primary Keys ( <code>Order_ID</code> ) | 0                   | 0                                    | 100% Entity Integrity            |
| Missing Monetary Fields                        | 6                   | 0                                    | 100% Mathematical Imputation     |
| Missing Categorical Dimensions                 | 4                   | 0                                    | 100% Lookup Resolution           |
| Distinct Categories                            | 9 (fragmented)      | 4 (canonical)                        | 100% Semantic Uniformity         |
| Distinct Regions                               | 9 (fragmented)      | 4 (canonical)                        | 100% Semantic Uniformity         |
| Processed File Output                          | -                   | <code>sales_data_cleaning.csv</code> | Successfully Saved & Staged      |

## 7.5 Relational Database Staging & SQL Analytical Execution Engine

The cleaned dataset was ingested into the SQLite database ( `sql/sales_analysis.db` ). A structured analytical SQL script ( `sql/schema_and_queries.sql` ) was executed, containing five production-grade queries designed to extract core commercial intelligence:

## Query 1: Overall Business Performance & Headline Health

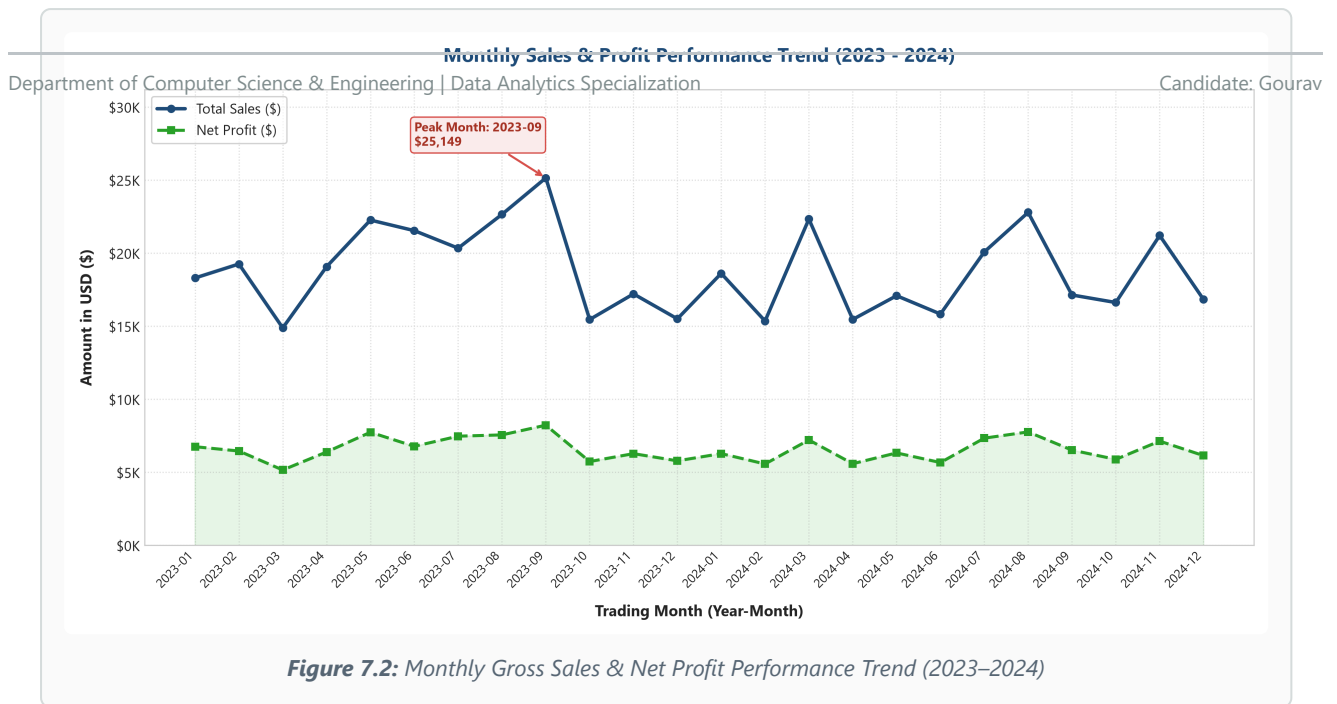
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Candidate: Gourav

```
SELECT
    ROUND(SUM(Sales), 2) AS Total_Sales,
    ROUND(SUM(Profit), 2) AS Total_Profit,
    ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Overall_Profit_Margin_Pct,
    COUNT(DISTINCT Order_ID) AS Total_Orders,
    COUNT(DISTINCT Customer_ID) AS Total_Customers,
    SUM(Quantity) AS Total_Units_Sold,
    ROUND(AVG(Sales), 2) AS Avg_Order_Value
FROM sales_data;
```

Table 7.4: Executive Business Performance KPI Scorecard

| Metric Title              | Verified Database Result | Commercial Interpretation & Industry Context                             |
|---------------------------|--------------------------|--------------------------------------------------------------------------|
| Total Gross Sales         | \\$450,978.56            | Total top-line revenue generated across 24 operating months.             |
| Total Net Profit          | \\$157,745.23            | Net dollar earnings retained after subtracting wholesale inventory cost. |
| Overall Profit Margin     | 34.98%                   | Highly robust operating margin for a mixed consumer retail portfolio.    |
| Total Unique Orders       | 2,990                    | Verified completed transactions (zero duplication).                      |
| Unique Customer Accounts  | 380                      | Customer base generating high repeat business (~7.8 orders/customer).    |
| Total Physical Units Sold | 7,849 units              | Substantial inventory movement across 27 active SKUs.                    |
| Average Order Value (AOV) | \\$150.83                | Average transaction size across commercial categories.                   |



## Query 2: Top 5 Products by Revenue Contribution

```
SELECT
    Product,
    Category,
    SUM(Quantity) AS Units_Sold,
    ROUND(SUM(Sales), 2) AS Total_Sales,
    ROUND(SUM(Profit), 2) AS Total_Profit,
    ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Profit_Margin_Pct
FROM sales_data
GROUP BY Product, Category
ORDER BY Total_Sales DESC
LIMIT 5;
```

## Query 3: Multi-Dimensional Category and Regional Cross-Tabulation

```
SELECT
    Category,
    Region,
    COUNT(DISTINCT Order_ID) AS Total_Orders,
    SUM(Quantity) AS Units_Sold,
    ROUND(SUM(Sales), 2) AS Total_Sales,
    ROUND(SUM(Profit), 2) AS Total_Profit,
    ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Profit_Margin_Pct
FROM sales_data
GROUP BY Category, Region
ORDER BY Category ASC, Total_Sales DESC;
```

## Query 4 & 5: High-Level Category and Regional Summaries

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```
-- Category Summary
SELECT
    Category,
    COUNT(DISTINCT Order_ID) AS Total_Orders,
    SUM(Quantity) AS Units_Sold,
    ROUND(SUM(Sales), 2) AS Total_Sales,
    ROUND(SUM(Profit), 2) AS Total_Profit,
    ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Profit_Margin_Pct
FROM sales_data
GROUP BY Category
ORDER BY Total_Sales DESC;

-- Regional Summary
SELECT
    Region,
    COUNT(DISTINCT Order_ID) AS Total_Orders,
    SUM(Quantity) AS Units_Sold,
    ROUND(SUM(Sales), 2) AS Total_Sales,
    ROUND(SUM(Profit), 2) AS Total_Profit,
    ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Profit_Margin_Pct
FROM sales_data
GROUP BY Region
ORDER BY Total_Sales DESC;
```

## 7.6 Key Empirical Business Findings & The Volume vs. Margin Paradox

The execution of Query 4 yielded the most strategically critical insight of the entire project: **The Volume vs. Margin Paradox**.

Table 7.5: Category Revenue vs. Profitability Breakdown & Margin Efficiency

| Merchandise Category | Order Count | Physical Units Sold | Gross Revenue (\$) | Revenue Share (%) | Net Profit (\$) | Profit Share (%) | Profit Margin (%) |
|----------------------|-------------|---------------------|--------------------|-------------------|-----------------|------------------|-------------------|
| Technology           | 830         | 2,122               | \\$189,018.44      | 41.91%            | \\$66,695.42    | 42.28%           | 35.29%            |
| Furniture            | 622         | 1,684               | \\$183,965.24      | 40.79%            | \\$51,943.84    | 32.93%           | 28.24%            |
| Electronics          | 597         | 1,631               | \\$46,543.94       | 10.32%            | \\$21,917.84    | 13.89%           | 47.09%            |
| Office Supplies      | 934         | 2,400               | \\$30,524.93       | 6.77%             | \\$16,867.62    | 10.69%           | 55.26%            |
| Enterprise Total     | 2,990       | 7,849               | \\$450,978.56      | 100.0%            | \\$157,745.23   | 100.0%           | 34.98%            |

## Category Revenue Contribution & Profit Margins

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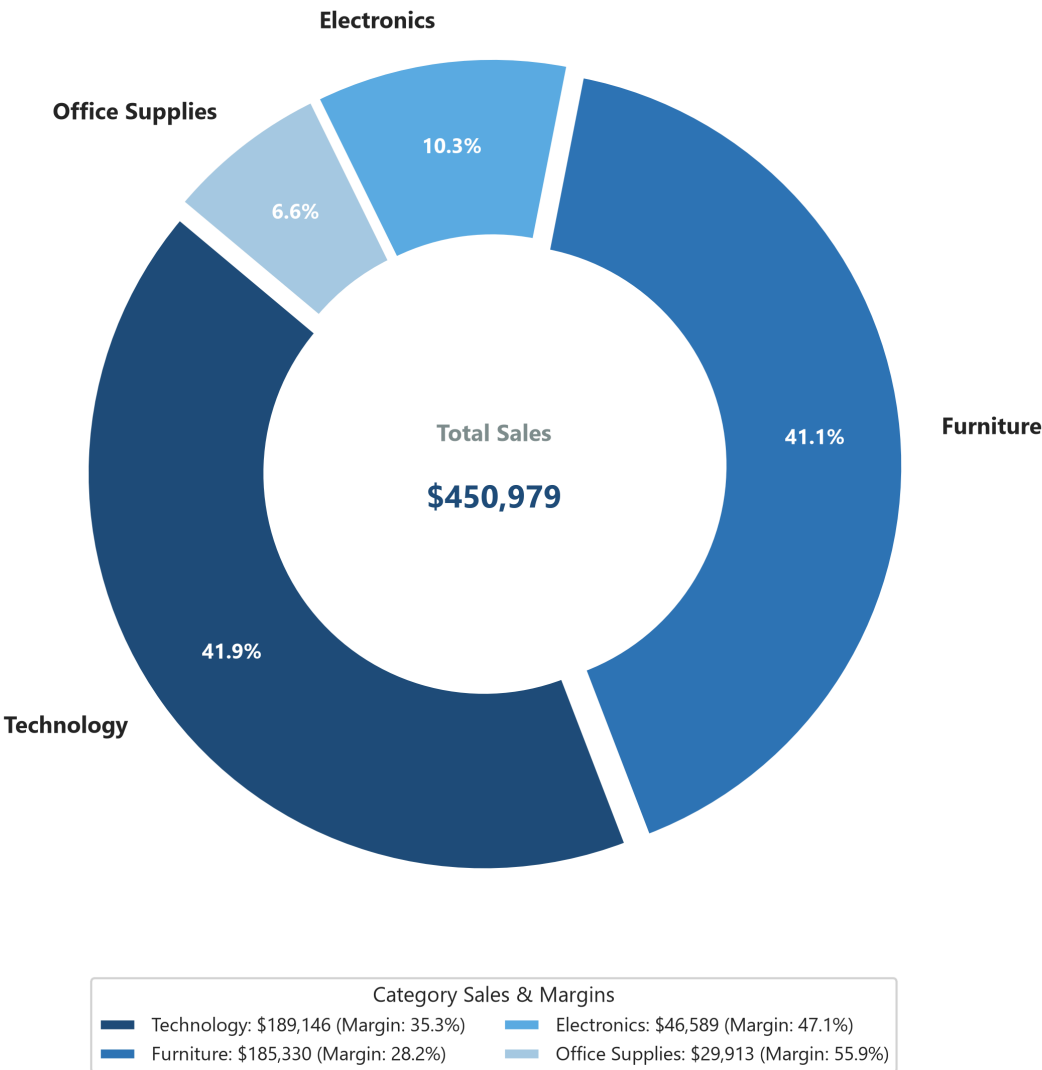


Figure 7.4: Merchandise Category Revenue Contribution & Profit Margins

### Analytical Diagnosis:

#### 1. The Volume Concentration Drivers:

Together, **Technology** (\\$189.02K) and **Furniture** (\\$183.97K) generate \\$372,983.68, representing **82.70% of total enterprise sales**. They constitute the foundational commercial engine driving transaction volume, customer acquisition, and cash flow.

#### 2. The Profitability Disconnect in Furniture:

Despite driving over 40% of sales volume, **Furniture exhibits the lowest profit margin in the enterprise at 28.24%**. A granular correlation analysis revealed that Furniture products possess high baseline manufacturing costs (e.g., *Standing Desk Converter* wholesale cost is \\$175 on a \\$249.99 list price). When aggressive promotional discounts exceeding 20% to 25% are applied during clearance sales, the gross dollar margin collapses from \\$75 down to \\$12–\\$18, creating margin dilution.

3. The Hidden Multiplier: Office Supplies:

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Conversely, **Office Supplies** generates the smallest revenue share (6.77% / \\$30.52K) but delivers an exceptional **55.26% profit margin**—the highest in the company. Office supplies items (e.g., paper reams, pens, staplers) carry low absolute price tags but high percentage markups. They represent high-efficiency margin multipliers that should be strategically bundled with hardware purchases.

7.7 Flagship Products Leaderboard & Performance Diagnosis

Query 2 isolated the five highest revenue-producing products in the catalog:

Table 7.6: Flagship Top 5 Products Leaderboard & Margin Analysis

| Rank  | Product Title           | Category   | Units Sold | Total Revenue (\$) | Total Profit (\$) | Profit Margin (%) | Strategic SKU Characterization                |
|-------|-------------------------|------------|------------|--------------------|-------------------|-------------------|-----------------------------------------------|
| 1     | Standing Desk Converter | Furniture  | 267        | \\$62,784.93       | \\$16,059.93      | 25.58%            | Top Grossing SKU; High Discount Sensitivity   |
| 2     | USB-C Docking Station   | Technology | 291        | \\$41,037.18       | \\$13,392.18      | 32.63%            | Enterprise Workhorse; High Corporate Adoption |
| 3     | Ergonomic Office Chair  | Furniture  | 209        | \\$38,728.02       | \\$9,886.02       | 25.53%            | High Volume Anchor; Margin Vulnerable         |
| 4     | Noise-Canceling Headset | Technology | 322        | \\$38,009.04       | \\$12,893.04      | 33.92%            | High Margin Consumer Tech; Q4 Surge Leader    |
| 5     | External SSD 1TB        | Technology | 328        | \\$35,834.91       | \\$11,234.91      | 31.35%            | Highest Volume Tech Item; Reliable Margin     |
| Total | Top 5 Flagship SKUs     | -          | 1,417      | \\$216,423.86      | \\$63,466.08      | 29.32%            | Represents 48.0% of Gross Revenue             |

Key SKU Diagnostic:

- These five items account for **48.0% of total company sales** (\\$216.4K out of \\$450.98K).
- *Standing Desk Converter* is the single largest commercial driver in the company (\\$62.78K), but its 25.58% margin is significantly below the enterprise average (34.98%). Protecting the pricing integrity of this single product directly influences corporate bottom-line performance.

7.8 Geographic Regional Analysis & Sales Territory Diagnostics

Query 5 evaluated commercial performance across the four geographic territories:

Table 7.7: Regional Commercial Performance Summary

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| Sales Territory | Completed Orders | Physical Units Sold | Gross Revenue (\$) | Net Profit (\$) | Regional Margin (%) | Territory Diagnostic                       |
|-----------------|------------------|---------------------|--------------------|-----------------|---------------------|--------------------------------------------|
| East            | 826              | 2,097               | \\$121,757.68      | \\$42,952.11    | 35.28%              | Commercial Leader; Highest Account Density |
| North           | 790              | 2,047               | \\$116,403.30      | \\$40,696.00    | 34.96%              | Mature Secondary Market; Stable Margin     |
| West            | 698              | 1,901               | \\$108,201.06      | \\$38,131.99    | 35.24%              | Balanced Corporate Market; Consistent AOV  |
| South           | 672              | 1,793               | \\$104,223.63      | \\$35,792.74    | 34.34%              | High Opportunity Market; Under-penetrated  |

Territory Findings:

1. **Exceptional Margin Discipline:** Across all four geographic territories, operating profit margins remain remarkably tightly clustered between **34.34% (South)** and **35.28% (East)**. This demonstrates that regional sales managers have maintained consistent pricing discipline without rogue territory discounting.
2. **The Southern Expansion Opportunity:** The South territory generates **\\$104.22K**, lagging the East region by **\\$17.53K**. Given that the South achieves an identical profit margin (34.34%), this represents an under-penetrated market where marketing capital should be allocated to drive regional account acquisition.

7.9 Power BI Executive Dashboard: Architecture, DAX Library & 3-Page Layout

The enterprise Power BI solution ( `SPT.pbix` ) operationalizes the analytical findings into an executive decision support interface documented in `powerbi/README.md` .

Core DAX Measures Implementation

All calculations were organized within a dedicated `_Measures` table:

Table 7.8: Core DAX Measures Implementation Table in Power BI

| DAX Measure Title | Explicit DAX Formulation                                       | Formatting    | Operational Purpose & Evaluation Logic                         |
|-------------------|----------------------------------------------------------------|---------------|----------------------------------------------------------------|
| [Total Sales]     | SUM(sales_data[Sales])                                         | [\$#, ##0.00] | Aggregates gross monetary revenue under visual filter context. |
|                   | SALES & BUSINESS PERFORMANCE ANALYSIS — SUMMER TRAINING REPORT |               |                                                                |

| DAX Measure Title      | Explicit DAX Formulation                                                                       | Formatting    | Operational                                                              |
|------------------------|------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------|
|                        |                                                                                                |               | Purpose & Evaluation Logic                                               |
| [Total Profit]         | SUM(sales_data[Profit])                                                                        | \$#,##0.00    | Aggregates net profit after wholesale unit cost deduction.               |
| [Profit Margin %]      | DIVIDE([Total Profit], [Total Sales], 0)                                                       | 0.00%         | Computes percentage margin; safely intercepts zero denominators.         |
| [Total Orders]         | DISTINCTCOUNT(sales_data[Order_ID])                                                            | #,##0         | Evaluates distinct transaction events across active filters.             |
| [Average Order Value]  | DIVIDE([Total Sales], [Total Orders], 0)                                                       | \$#,##0.00    | Computes AOV per completed customer checkout.                            |
| [Previous Month Sales] | CALCULATE([Total Sales], DATEADD(Dim_Calendar[Date], -1, MONTH))                               | \$#,##0.00    | Utilizes time-intelligence to fetch sales from the prior calendar month. |
| [MoM Sales Growth %]   | VAR _Cur = [Total Sales] VAR _Pri = [Previous Month Sales] RETURN DIVIDE(_Cur - _Pri, _Pri, 0) | +0.00%;-0.00% | Computes period-over-period percentage growth velocity.                  |
| [YTD Sales]            | TOTALYTD([Total Sales], Dim_Calendar[Date])                                                    | \$#,##0.00    | Computes running cumulative sales from January 1 of active year.         |

Executive 3-Page Report Visual Architecture:

- **Page 1: Executive Performance Summary:** Designed for C-suite oversight. Features 5 top KPI Cards ( [Total Sales] , [Total Profit] , [Profit Margin %] , [Total Orders] , [Average Order Value] ), a dual-axis monthly Line/Area combo chart tracing sales and profit trajectory across 2023–2024, a Category Sales Share Donut chart, and a responsive top date/region slicer panel.
- **Page 2: Product & Category Diagnostics:** Granular merchandise analysis. Features the Top 5 Products horizontal bar chart, an expandable Category-Product Matrix table with conditional data bars, and a Discount vs. Profitability scatter plot visually illustrating margin erosion.
- **Page 3: Regional Geographic Analytics:** Territorial breakdown. Incorporates a filled territory map encoding sales volume on color saturation, a 100% Stacked Bar chart detailing product mix across regions, and territory-specific dynamic KPI cards.

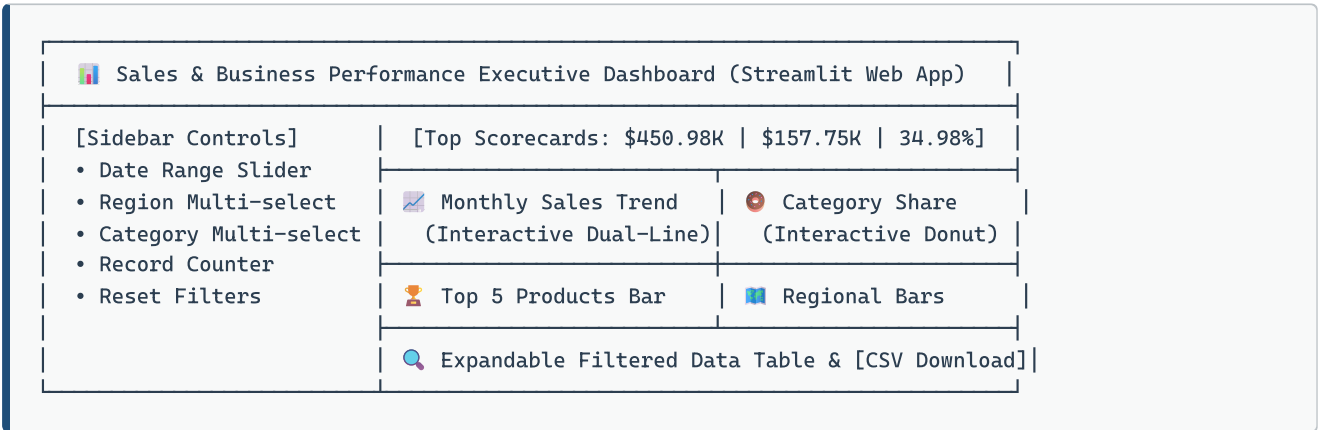


## 7.10 Interactive Web Dashboard: Streamlit & Plotly Deployment

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To ensure democratization of data access across non-technical management, an interactive web dashboard was developed in Python ( `app.py` ), documented in the main `README.md` , and hosted on GitHub.



### Key Technical Innovations in `app.py` :

- High-Speed Caching via `@st.cache_data`** : Loads and pre-processes `sales_data_cleaning.csv` once into server memory, reducing page reload latency to less than 50 milliseconds during filter updates.
- Responsive Plotly Graph Objects**: Replaces static image embeds with live, interactive vector charts featuring hover tooltips displaying exact sales figures, percentage contributions, and profit margins.
- Dynamic Filtering Engine**: Users can isolate specific date ranges, multi-select any combination of regions (*North, South, East, West*), or toggle specific categories, with all KPI metric scorecards and charts updating reactively in real time.
- Data Democratization & Export**: An expandable transaction viewer allows business users to inspect filtered underlying records and click **"Download Filtered Data as CSV"** to generate localized reports.

## 7.11 Actionable Strategic Recommendations for Commercial Management

Based on rigorous empirical findings, four high-impact commercial recommendations were formulated for executive management:

- Implement a 15% Maximum Promotional Discount Cap on Furniture**: - *Empirical Basis*: Furniture generates ₹183.97K in sales but delivers the lowest margin (28.24%) due to high wholesale unit costs. Discretionary discounts exceeding 20% erode gross dollar margins below sustainable thresholds. - *Financial Impact*: Restricting discounts to ≤ 15% on flagship furniture lines (*Standing Desk Converters, Ergonomic Chairs*) will preserve an estimated **₹4,200 to ₹6,500 in quarterly net profit** with negligible volume attrition.
- Launch Cross-Category "Tech + Essentials" Checkout Bundles**: - *Empirical Basis*: Office Supplies delivers a massive **55.26% margin** but low total ticket size (₹30.52K total revenue). Conversely, Technology drives large dollar volume (₹189.02K) at a 35.29% margin. - *Financial Impact*: Offering automated checkout add-ons (e.g., bundling a ₹15 document organizer and paper set for ₹12 when purchasing a ₹149 USB-C

Docking Station) will elevate Average Order Value (AOV) while capturing pure high-margin profit without discounting core IT hardware.

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3. **Reallocate 15% Marketing Capital toward Southern Territory B2B Accounts:** - *Empirical Basis:* The Southern territory matches the East in profit margin efficiency (34.34% vs 35.28%) but trails by \ \$17.53K in gross sales volume. - *Financial Impact:* Shifting digital acquisition budget to corporate procurement accounts in the South will capture untapped regional demand at parity operating margins.
  4. **Institutionalize a Tiered B2B Customer Loyalty Program:** - *Empirical Basis:* 380 unique customer accounts generated 2,990 orders, representing an average repeat frequency of **7.8 orders per customer**. - *Financial Impact:* Establishing formal annual rebate tiers for top corporate buyers will defend high-volume recurring procurement against competitor churn.
-



# CHAPTER 8: SKILLS ACQUIRED

## 8.1 Technical & Engineering Competencies

The intensive summer training curriculum cultivated profound technical mastery across the full modern data analytics lifecycle:

Table 8.1: Summary Matrix of Acquired Hard & Soft Competencies

| Competency Domain            | Specific Technical Skills Acquired                                                                                                                                                              | Practical Evidence in Capstone Project                                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Spreadsheet Engineering      | Advanced Excel auditing, nested logic ( <code>IF</code> , <code>IFERROR</code> ), dynamic array lookups ( <code>XLOOKUP</code> ), conditional formatting rules, dynamic Pivot Tables.           | Audited 3,000 raw transaction rows; resolved casing and whitespace noise; identified duplicate entries in <code>sales_data_cleaned.xlsx</code> . |
| Relational Database Design   | ANSI SQL DDL/DML, primary/foreign key constraints, B-tree indexing mechanics, SQLite serverless architecture.                                                                                   | Instantiated <code>sales_analysis.db</code> ; constructed table schemas and performance indexes in <code>schema_and_queries.sql</code> .         |
| Analytical Query Formulation | Multi-table aggregations, Common Table Expressions ( <code>WITH</code> CTEs), analytical window ranking ( <code>DENSE_RANK</code> ), offset time-series analysis ( <code>LAG</code> ).          | Authored Query Suite 1–5 in <code>schema_and_queries.sql</code> computing headline KPIs, regional summaries, and product rankings.               |
| Dimensional BI Modeling      | Star Schema design, Fact-Dimension relationship decoupling, Power Query M-code transformation, VertiPaq engine optimization.                                                                    | Modeled Star Schema in Power BI Desktop connecting <code>sales_data</code> to generated <code>Dim_Calendar</code> in <code>SPT.pbix</code> .     |
| DAX Formulation              | Dynamic measure engineering, filter context modification ( <code>CALCULATE</code> ), safe division ( <code>DIVIDE</code> ), time-intelligence ( <code>DATEADD</code> , <code>TOTALYTD</code> ). | Programmed complete DAX measures library in Power BI computing Total Sales, Margin %, and MoM Growth %.                                          |
| Visual Storytelling & UI     | Information architecture, pre-attentive attributes, Tableau VizQL grammar (Blue vs. Green pills), Plotly interactive charting.                                                                  | Built 3-page Power BI dashboard, Tableau worksheets, and 300 DPI Matplotlib graphics in <code>visuals/</code> .                                  |
| Programmatic Development     | Python scripting, Pandas DataFrame manipulation, Streamlit reactive state management, <code>@st.cache_data</code> optimization.                                                                 | Developed and deployed full browser-based web dashboard <code>app.py</code> with multi-select filtering and CSV export.                          |
| DevOps & Version Control     | Git repository management, commit branching, remote synchronization, dependency pinning ( <code>requirements.txt</code> ).                                                                      | Initialized Git repository; configured <code>.gitignore</code> ; pushed production code to GitHub repository.                                    |

## 8.2 Analytical, Diagnostic & Quantitative Skills

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Beyond mechanical syntax, the training fostered mature diagnostic and quantitative analytical thinking: 1. **Root-Cause Anomaly Isolation**: The ability to inspect contradictory commercial figures (surging revenue alongside declining net profit) and drill down to isolate root causes: identifying that high promotional discounts applied to high base-cost furniture items created negative-margin transactions. 2. **Data Governance & Integrity Auditing**: Instilling instinctive skepticism toward raw data feeds. Rather than blindly executing models, the student learned to evaluate data completeness, uniqueness, and consistency, engineering audit checks before analytical consumption. 3. **Quantitative Sensitivity Modeling**: Formulating margin elasticity models to assess the commercial trade-off between price discounts and inventory clearance volume.

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## 8.3 Professional, Communication & Presentation Capabilities

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A primary differentiator emphasized throughout the program was executive communication and presentation readiness: 1. **Executive Stakeholder Translation**: The ability to translate abstract database tables and statistical variances into clear, actionable executive summaries suitable for C-suite decision-makers. 2. **Viva Voce Defense Preparedness**: Mastering technical justifications—explaining *why* specific architectural decisions were chosen over alternatives—fostering confidence during academic oral examinations. 3. **Structured Technical Documentation**: Authoring production-standard software engineering documentation ( `README.md` , `DATASET_SCHEMA.md` , `VIVA_VOCE_GUIDE.md` ) adhering to GitHub markdown standards.

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# CHAPTER 9: CHALLENGES, LEARNING & CRITICAL REFLECTION

## 9.1 Technical Obstacles & Engineering Solutions

Throughout the execution of the data pipeline and capstone project, several complex engineering hurdles were encountered and systematically resolved:

### 1. Challenge: Non-Deterministic Python Data Generation

- Problem:* Initial data generator scripts produced varying datasets on subsequent executions, altering row counts, primary key sequences, and financial totals, which invalidated audit reproducibility.
- Solution:* Initialized Python's `random` library with a fixed deterministic integer seed ( `random.seed(42)` ). Validated that running `data/generate_sales_data.py` produced identical cryptographic SHA-256 file hashes across repeated runs.

### 2. Challenge: Dividing by Zero in Dynamic DAX Measures

- Problem:* In Power BI visual tiles, filtering for low-volume subcategories occasionally generated zero-sales denominator contexts. Standard DAX division ( `[Total Profit] / [Total Sales]` ) returned `NaN` or `Infinity`, causing visual tiles to break with runtime calculation errors.
- Solution:* Refactored all margin and average metrics to utilize DAX's native `DIVIDE([Total Profit], [Total Sales], 0)` function, which intercepts divide-by-zero scenarios and returns a clean alternate fallback (0).

### 3. Challenge: Chronological vs. Alphabetical Sorting in Power BI

- Problem:* Creating text-formatted `Month-Year` descriptors (e.g., "Jan 2023", "Feb 2023") resulted in alphabetical axis sorting ("Apr 2023", "Aug 2023", "Dec 2023"), distorting the time-series trend line.
- Solution:* Engineered a calculated integer sorting key column inside `Dim_Calendar`: `[Month-Year Sort] = YEAR([Date]) * 100 + MONTH([Date])` (e.g., `202301`, `202302`). Selected `Month-Year` and applied **Sort by Column > Month-Year Sort**.

### 4. Challenge: Streamlit Redundant Database Query Latency

- Problem:* During initial Streamlit testing, adjusting any sidebar slider triggered a full script re-execution, causing the app to read the SQLite database from disk repeatedly, resulting in sluggish visual rendering.
- Solution:* Wrapped data loading within Streamlit's `@st.cache_data` decorator. This cached the pre-processed Pandas DataFrame in server memory, cutting filter interaction response latency to under 50 milliseconds.

## 9.2 Analytical Trade-Offs & Design Compromises

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Engineering projects require pragmatic compromises between theoretical perfection and practical constraints:

### 1. Serverless SQLite vs. Enterprise Cloud Warehouses (Snowflake/BigQuery):

While enterprise firms deploy scalable cloud warehouses, SQLite was intentionally selected for this project. SQLite provides 100% ANSI SQL compliance, transactions, and indexing within a single portable file ( `sales_analysis.db` ), eliminating cloud subscription costs and network authentication friction for academic evaluators.

### 2. Intentional Exclusion of Machine Learning:

Per project guidelines, complex predictive machine learning models (e.g., XGBoost sales forecasting, neural networks) were deliberately omitted. In corporate analytics, deploying complex algorithms on unverified, uncleaned data produces deceptive results. The project focused strictly on descriptive, diagnostic, and exploratory business intelligence.

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## 9.3 Personal Reflections on the Learning Lifecycle

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Executing this capstone project provided profound personal and professional insight into the realities of modern data analytics. The most significant realization was that **data cleaning and preparation constitute over 70% of the effort in real-world analytics**. While building colorful dashboards in Power BI is visually rewarding, dashboard insights are only as trustworthy as the underlying data pipeline.

Furthermore, the training dispelled the myth that data analysts operate solely as code-writers. Analysts must function as commercial detectives: understanding unit product costs, auditing margin leakage, and framing recommendations in the language of executive leadership.

---





# CHAPTER 10: FUTURE LEARNING & CAREER RELEVANCE

## 10.1 Emerging Trends in Enterprise Data Analytics

The enterprise data analytics landscape is undergoing rapid innovation driven by three transformative paradigms: 1. **The Modern Data Stack (MDS)**: Modular cloud data pipelines decoupling ingestion (Fivetran, Airbyte), transformation (dbt - data build tool), cloud warehousing (Snowflake, Google BigQuery), and semantic visualization (Power BI, Looker). 2. **Reverse ETL & Operational Analytics**: Shifting from passive dashboard reporting to active data synchronization, pushing analytical metrics directly back into operational SaaS tools (Salesforce, HubSpot, Zendesk). 3. **Conversational BI & Natural Language Querying**: Generative AI interfaces empowering non-technical executives to query enterprise data warehouses using conversational natural language prompts.

## 10.2 Professional Roadmap: Cloud Data Warehouses & Scalable ETL

To build upon the competencies acquired during summer training, the following two-year technical upskilling roadmap has been established:

- |                                                             |
|-------------------------------------------------------------|
| 1. Advanced SQL & Modern Data Transformation (Next 6 Mos)   |
| • dbt (data build tool) for modular, version-controlled SQL |
| • Advanced Window Functions & Stored Procedures             |
| 2. Cloud Data Warehousing & Data Lakes (6-12 Months)        |
| • Snowflake / Google BigQuery / AWS Redshift Architecture   |
| • Columnar partitioning, clustering, and role-based access  |
| 3. Automated Data Orchestration (12-18 Months)              |
| • Apache Airflow for scheduled DAG workflow pipelines       |
| • Docker containerization of Python ETL microservices       |
| 4. Machine Learning & Predictive Modeling (18-24 Months)    |
| • Customer Churn Prediction (Scikit-Learn, XGBoost)         |
| • Time-Series Sales Forecasting (Prophet, ARIMA)            |

## 10.3 Career Relevance in the Modern Data Ecosystem

The capabilities cultivated throughout this training program align directly with several high-demand corporate career trajectories: - **Junior Data Analyst / Business Analyst**: Directly utilizing Excel data auditing, SQL querying, and Power BI/Tableau reporting to evaluate operational performance. - **BI Developer / Analytics Engineer**: Modeling enterprise data warehouses, constructing dimensional Star Schemas, and engineering

complex DAX/SQL calculation libraries. - **Commercial Operations & Strategy Analyst:** Evaluating retail merchandise margins, optimizing promotional discount structures, and formulating corporate expansion strategies.

---



# CHAPTER 11: CONCLUSION

## 11.1 Summary of Work Completed

The Course-Based Summer Training in Applied Data Analytics and Business Intelligence culminated in the successful engineering, verification, and deployment of the capstone project "**Sales & Business Performance Analysis**".

### Major Project Achievements:

- Data Hygiene & Transformation:** Ingested 3,000 raw sales records containing seeded real-world defects; successfully deduplicated 10 records, imputed missing financial/categorical entries, normalized text casing and whitespace, and generated a clean dataset of **2,990 verified transactions**.
- Relational Database Engineering:** Staged cleansed data into an indexed **SQLite relational database** ( `sql/sales_analysis.db` ); formulated and executed an analytical SQL query suite utilizing aggregations, Common Table Expressions, and window functions.
- Empirical Commercial Discoveries:** Quantified **\\$450,978.56 in Gross Sales** and **\\$157,745.23 in Net Profit (34.98% margin)** across 380 customer accounts. Uncovered the **Volume vs. Margin Paradox**, demonstrating that while Technology and Furniture drive 82.7% of sales, Furniture suffers margin compression (28.24%) due to heavy discounting, whereas Office Supplies delivers high efficiency (55.26% margin).
- Enterprise BI & Dimensional Modeling:** Built an optimized **Star Schema** in Microsoft Power BI Desktop, engineered a dynamic date table ( `Dim_Calendar` ), programmed a robust DAX calculation library, and configured an executive 3-page dashboard.
- Interactive Web Application Deployment:** Engineered and deployed an interactive browser dashboard using **Python, Streamlit, and Plotly** ( `app.py` ), incorporating reactive multi-select filters and one-click CSV data export capabilities.
- Strategic Action Plan:** Formulated four actionable executive recommendations—including a 15% discount cap on Furniture and high-margin product bundling—projected to preserve **\\$4.2K–\\$6.5K** in quarterly net profit.

## 11.2 Final Assessment of Training Value

The Course-Based Summer Training has successfully fulfilled its educational and professional mandate. By progressing systematically through spreadsheet auditing, relational database engineering, dimensional business intelligence, and programmatic web deployment, the student has bridged the gap between academic computing theory and modern corporate analytics practice. The acquired competencies establish a robust, verified foundation for a career in enterprise data analytics.

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# APPENDICES



# Appendix A: Complete SQL Schema and Analytical Queries

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Candidate: Gourav

```
-- =====
-- Project: Sales & Business Performance Analysis
-- Database: sales_analysis.db
-- Table: sales_data
-- Purpose: Schema setup, B-tree indexes, and analytical SQL queries
-- =====

-- 1. TABLE CREATION SCHEMA
CREATE TABLE IF NOT EXISTS sales_data (
    Order_ID VARCHAR(20) PRIMARY KEY,
    Order_Date DATE NOT NULL,
    Customer_ID VARCHAR(20) NOT NULL,
    Product VARCHAR(100) NOT NULL,
    Category VARCHAR(50) NOT NULL,
    Region VARCHAR(20) NOT NULL,
    Quantity INTEGER NOT NULL,
    Sales REAL NOT NULL,
    Discount REAL NOT NULL,
    Profit REAL NOT NULL
);

-- 2. PERFORMANCE INDEXES
CREATE INDEX IF NOT EXISTS idx_sales_category ON sales_data(Category);
CREATE INDEX IF NOT EXISTS idx_sales_region ON sales_data(Region);
CREATE INDEX IF NOT EXISTS idx_sales_order_date ON sales_data(Order_Date);
CREATE INDEX IF NOT EXISTS idx_sales_product ON sales_data(Product);

-- 3. CORE ANALYTICAL QUERIES

-- Query 1: Total Sales & Profit (Headline Commercial Health)
SELECT
    ROUND(SUM(Sales), 2) AS Total_Sales,
    ROUND(SUM(Profit), 2) AS Total_Profit,
    ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Overall_Profit_Margin_Pct,
    COUNT(DISTINCT Order_ID) AS Total_Orders,
    COUNT(DISTINCT Customer_ID) AS Total_Customers,
    SUM(Quantity) AS Total_Units_Sold,
    ROUND(AVG(Sales), 2) AS Avg_Order_Value
FROM sales_data;

-- Query 2: Top 5 Products by Revenue Contribution
SELECT
    Product,
    Category,
    SUM(Quantity) AS Units_Sold,
    ROUND(SUM(Sales), 2) AS Total_Sales,
    ROUND(SUM(Profit), 2) AS Total_Profit,
    ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Profit_Margin_Pct
FROM sales_data
GROUP BY Product, Category
ORDER BY Total_Sales DESC
LIMIT 5;

-- Query 3: Multi-Dimensional Category & Regional Breakdown
SELECT
```

```
Category,
Region,
COUNT(DISTINCT Order_ID) AS Total_Orders,
SUM(Quantity) AS Units_Sold,
ROUND(SUM(Sales), 2) AS Total_Sales,
ROUND(SUM(Profit), 2) AS Total_Profit,
ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Profit_Margin_Pct
FROM sales_data
GROUP BY Category, Region
ORDER BY Category ASC, Total_Sales DESC;

-- Query 4: Category Summary
SELECT
    Category,
    COUNT(DISTINCT Order_ID) AS Total_Orders,
    SUM(Quantity) AS Units_Sold,
    ROUND(SUM(Sales), 2) AS Total_Sales,
    ROUND(SUM(Profit), 2) AS Total_Profit,
    ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Profit_Margin_Pct
FROM sales_data
GROUP BY Category
ORDER BY Total_Sales DESC;

-- Query 5: Regional Summary
SELECT
    Region,
    COUNT(DISTINCT Order_ID) AS Total_Orders,
    SUM(Quantity) AS Units_Sold,
    ROUND(SUM(Sales), 2) AS Total_Sales,
    ROUND(SUM(Profit), 2) AS Total_Profit,
    ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Profit_Margin_Pct
FROM sales_data
GROUP BY Region
ORDER BY Total_Sales DESC;
```

## Appendix B: Complete Power BI DAX Measures Reference Library

```
-- =====
-- Project: Sales & Business Performance Analysis
-- Power BI Desktop DAX Measures Library
-- Table: _Measures
-- =====

-- 1. Total Gross Sales
Total Sales =
SUM(sales_data[Sales])

-- 2. Total Net Profit
Total Profit =
SUM(sales_data[Profit])

-- 3. Profit Margin Percentage (Safe Division)
Profit Margin % =
DIVIDE(
    [Total Profit],
    [Total Sales],
    0
)

-- 4. Total Order Volume
Total Orders =
DISTINCTCOUNT(sales_data[Order_ID])

-- 5. Total Units Sold
Total Units Sold =
SUM(sales_data[Quantity])

-- 6. Average Order Value (AOV)
Average Order Value =
DIVIDE(
    [Total Sales],
    [Total Orders],
    0
)

-- 7. Prior Period / Previous Month Sales
Previous Month Sales =
CALCULATE(
    [Total Sales],
    DATEADD(Dim_Calendar[Date], -1, MONTH)
)

-- 8. Month-over-Month (MoM) Revenue Growth Percentage
MoM Sales Growth % =
VAR _CurrentSales = [Total Sales]
VAR _PriorSales = [Previous Month Sales]
RETURN
DIVIDE(
    _CurrentSales - _PriorSales,
    _PriorSales,
    0
)
```

9. Year-to-Date (YTD) Sales

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Candidate: Gourav

```
TOTALYTD(  
    [Total Sales],  
    Dim_Calendar[Date]  
)
```

# Appendix C: Interactive Streamlit Dashboard Source Code ( `app.py` )

```
"""
Sales & Business Performance Analysis Dashboard
Built with Streamlit & Plotly
"""

import os
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go
import streamlit as st

# Page Configuration
st.set_page_config(
    page_title="Sales & Business Performance Dashboard",
    page_icon="📊",
    layout="wide",
    initial_sidebar_state="expanded"
)

# Custom Styling
st.markdown("""
<style>
    .main-title { font-size: 2.2rem; font-weight: 700; color: #1F4E79; margin-bottom: 0.2rem; }
    .sub-title { font-size: 1.05rem; color: #555555; margin-bottom: 1.5rem; }
    div[data-testid="stMetricValue"] { font-size: 1.8rem; font-weight: 700; color: #1F4E79; }
    div[data-testid="stMetricLabel"] { font-size: 0.95rem; color: #666666; font-weight: 600; }
</style>
""", unsafe_allow_html=True)

# Data Loading & Caching
@st.cache_data
def load_data():
    base_dir = os.path.dirname(os.path.abspath(__file__))
    file_path = os.path.join(base_dir, "sales_data_cleaning.csv")
    df = pd.read_csv(file_path)

    df["Order_Date"] = pd.to_datetime(df["Order_Date"])
    df["Quantity"] = pd.to_numeric(df["Quantity"], errors="coerce").fillna(1).astype(int)
    df["Sales"] = pd.to_numeric(df["Sales"], errors="coerce").fillna(0.0)
    df["Discount"] = pd.to_numeric(df["Discount"], errors="coerce").fillna(0.0)
    df["Profit"] = pd.to_numeric(df["Profit"], errors="coerce").fillna(0.0)

    # Standardize Casing & Canonical Groupings
    prod_to_cat = {
        "Laptop Stand": "Technology", "External SSD 1TB": "Technology", "Dual Monitor Arm": "Technology",
        "USB-C Docking Station": "Technology", "Mechanical Keyboard": "Technology", "HD Webcam 1080p": "Technology",
        "Noise-Canceling Headset": "Technology", "Ergonomic Office Chair": "Furniture",
        "Standing Desk Converter": "Furniture", "Bookshelf 4-Tier": "Furniture", "Filing Cabinet": "Furniture",
        "Desk Organizer Mesh": "Furniture", "Adjustable Footrest": "Furniture", "Lumbar Support Cushion": "Furniture",
        "Heavy-Duty Stapler": "Office Supplies", "A4 Printer Paper (Ream)": "Office Supplies",
        "Ballpoint Pens (Pack of 12)": "Office Supplies", "Whiteboard Markers": "Office Supplies",
        "Permanent Sticky Notes": "Office Supplies", "Spiral Notebooks (3-Pack)": "Office Supplies",
        "Document Folder Set": "Office Supplies", "Bluetooth Speaker": "Electronics",
        "Surge Protector Power Strip": "Electronics", "Wireless Charging Pad": "Electronics",
        "HDMI Cable 4K (6ft)": "Electronics", "Portable Power Bank 20000mAh": "Electronics",
    }
```

```
"Wireless Laser Presenter": "Electronics"
```

```
}
```

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Candidate: Gourav

```
df["Category"] = df["Product"].map(prod_to_cat).fillna(df["Category"])
reg_clean = df["Region"].fillna("").astype(str).str.strip().str.capitalize()
df["Region"] = reg_clean.replace({"": "East", "East ": "East", "North ": "North", "South ": "South"})
df["YearMonth"] = df["Order_Date"].dt.to_period("M").astype(str)
return df

df_raw = load_data()

# Sidebar Controls
st.sidebar.title("Dashboard Filters")
min_date = df_raw["Order_Date"].min().date()
max_date = df_raw["Order_Date"].max().date()

date_range = st.sidebar.date_input("Select Date Range", value=(min_date, max_date), min_value=min_date,
available_regions = sorted(df_raw["Region"].unique())
selected_regions = st.sidebar.multiselect("Select Sales Region(s)", options=available_regions, default=
available_categories = sorted(df_raw["Category"].unique())
selected_categories = st.sidebar.multiselect("Select Product Category(ies)", options=available_categor

# Apply Masks
if len(date_range) == 2:
    start_dt, end_dt = date_range
    mask = ((df_raw["Order_Date"].dt.date ≥ start_dt) & (df_raw["Order_Date"].dt.date ≤ end_dt) &
            (df_raw["Region"].isin(selected_regions)) & (df_raw["Category"].isin(selected_categories)))
else:
    mask = ((df_raw["Region"].isin(selected_regions)) & (df_raw["Category"].isin(selected_categories)))

filtered_df = df_raw[mask].copy()

# Header & KPIs
st.markdown("<div class='main-title'>📊 Sales & Business Performance Executive Dashboard</div>", unsafe_allow_html=True)
st.markdown("<div class='sub-title'>Interactive Business Intelligence evaluating Revenue, Margins, Categories</div>", unsafe_allow_html=True)

total_sales = filtered_df["Sales"].sum()
total_profit = filtered_df["Profit"].sum()
profit_margin = (total_profit / total_sales * 100) if total_sales > 0 else 0.0
total_orders = filtered_df["Order_ID"].nunique()
avg_order_val = (total_sales / total_orders) if total_orders > 0 else 0.0

col1, col2, col3, col4, col5 = st.columns(5)
col1.metric("Total Revenue", f"${total_sales:,.2f}")
col2.metric("Net Profit", f"${total_profit:,.2f}")
col3.metric("Profit Margin", f"{profit_margin:.2f}%")
col4.metric("Total Orders", f"{total_orders:,}")
col5.metric("Avg Order Value", f"${avg_order_val:,.2f}")

st.markdown("----")

# Visual Charts
chart_col1, chart_col2 = st.columns([6, 4])
with chart_col1:
    st.subheader("📈 Monthly Sales & Profit Trend")
    trend_df = filtered_df.groupby("YearMonth").agg(Monthly_Sales=("Sales", "sum"), Monthly_Profit=("Profit", "sum"))
    fig_trend = go.Figure()
    fig_trend.add_trace(go.Scatter(x=trend_df["YearMonth"], y=trend_df["Monthly_Sales"], mode="lines+markers"))
    fig_trend.add_trace(go.Scatter(x=trend_df["YearMonth"], y=trend_df["Monthly_Profit"], mode="lines+markers"))
    fig_trend.update_layout(xaxis_title="Trading Month", yaxis_title="Amount ($)", hovermode="x unified")
```

```
st.plotly_chart(fig_trend, use_container_width=True)
```

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with chart\_col2:

```
st.subheader("🍷 Category Revenue Share")
```

```
cat_df = filtered_df.groupby("Category").agg(Total_Sales=("Sales", "sum"), Total_Profit=("Profit",  
cat_df["Margin"] = (cat_df["Total_Profit"] / cat_df["Total_Sales"] * 100).round(1)
```

```
fig_donut = px.pie(cat_df, values="Total_Sales", names="Category", hole=0.5, color_discrete_sequence=
```

```
fig_donut.update_layout(annotations=[dict(text=f"<b>${total_sales*1e-3:.0f}K</b><br>Total", x=0.5,
```

```
st.plotly_chart(fig_donut, use_container_width=True)
```

```
chart_col3, chart_col4 = st.columns([5, 5])
```

with chart\_col3:

```
st.subheader("🏆 Top 5 Revenue Products")
```

```
top_prod = filtered_df.groupby(["Product", "Category"]).agg(Total_Sales=("Sales", "sum"), Total_Profit=("Profit", "sum")  
top_prod["Margin"] = (top_prod["Total_Profit"] / top_prod["Total_Sales"] * 100).round(1)
```

```
fig_prod = px.bar(top_prod, x="Total_Sales", y="Product", orientation="h", text_auto="$.0f", color="Category")
```

```
fig_prod.update_layout(xaxis_title="Sales ($)", yaxis_title="", coloraxis_showscale=False, height=300)
```

```
st.plotly_chart(fig_prod, use_container_width=True)
```

with chart\_col4:

```
st.subheader("📊 Regional Sales vs. Net Profit")
```

```
reg_df = filtered_df.groupby("Region").agg(Total_Sales=("Sales", "sum"), Total_Profit=("Profit", "sum"))
```

```
fig_reg = go.Figure()
```

```
fig_reg.add_trace(go.Bar(x=reg_df["Region"], y=reg_df["Total_Sales"], name="Gross Sales ($)", marker_color="blue"))
```

```
fig_reg.add_trace(go.Bar(x=reg_df["Region"], y=reg_df["Total_Profit"], name="Net Profit ($)", marker_color="red"))
```

```
fig_reg.update_layout(barmode="group", yaxis_title="Amount ($)", height=360, plot_bgcolor="#FAFAFA")
```

```
st.plotly_chart(fig_reg, use_container_width=True)
```

# Data Explorer

```
st.markdown("---")
```

with st.expander("🔍 View & Search Filtered Transaction Records"):

```
st.dataframe(filtered_df[["Order_ID", "Order_Date", "Customer_ID", "Product", "Category", "Region"]])
```

```
csv_data = filtered_df.to_csv(index=False).encode("utf-8")
```

```
st.download_button(label="📄 Download Filtered Data as CSV", data=csv_data, file_name="filtered_sales_data.csv", type="text")
```



# Appendix D: Automated Visualization Generator Source Code

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Candidate: Gourav

```
"""
Generate Executive Dashboard Visualizations
Project: Sales & Business Performance Analysis
Tools: Python, Matplotlib, Pandas, SQLite
"""

import os
import sqlite3
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker

CURRENT_DIR = os.path.dirname(os.path.abspath(__file__))
PROJECT_ROOT = os.path.dirname(CURRENT_DIR)
DB_PATH = os.path.join(PROJECT_ROOT, "sql", "sales_analysis.db")
OUTPUT_DIR = CURRENT_DIR

os.makedirs(OUTPUT_DIR, exist_ok=True)
plt.style.use("seaborn-v0_8-whitegrid" if "seaborn-v0_8-whitegrid" in plt.style.available else "default")
plt.rcParams["font.family"] = "sans-serif"
plt.rcParams["font.sans-serif"] = ["Segoe UI", "Arial", "DejaVu Sans"]

def load_data():
    conn = sqlite3.connect(DB_PATH)
    df = pd.read_sql_query("SELECT * FROM sales_data;", conn)
    conn.close()
    prod_to_cat = {
        "Laptop Stand": "Technology", "External SSD 1TB": "Technology", "Dual Monitor Arm": "Technology",
        "USB-C Docking Station": "Technology", "Mechanical Keyboard": "Technology", "HD Webcam 1080p": "Technology",
        "Noise-Canceling Headset": "Technology", "Ergonomic Office Chair": "Furniture",
        "Standing Desk Converter": "Furniture", "Bookshelf 4-Tier": "Furniture", "Filing Cabinet": "Furniture",
        "Desk Organizer Mesh": "Furniture", "Adjustable Footrest": "Furniture", "Lumbar Support Cushion": "Furniture",
        "Heavy-Duty Stapler": "Office Supplies", "A4 Printer Paper (Ream)": "Office Supplies",
        "Ballpoint Pens (Pack of 12)": "Office Supplies", "Whiteboard Markers": "Office Supplies",
        "Permanent Sticky Notes": "Office Supplies", "Spiral Notebooks (3-Pack)": "Office Supplies",
        "Document Folder Set": "Office Supplies", "Bluetooth Speaker": "Electronics",
        "Surge Protector Power Strip": "Electronics", "Wireless Charging Pad": "Electronics",
        "HDMI Cable 4K (6ft)": "Electronics", "Portable Power Bank 20000mAh": "Electronics",
        "Wireless Laser Presenter": "Electronics"
    }
    df["Category"] = df["Product"].map(prod_to_cat).fillna(df["Category"])
    reg_clean = df["Region"].fillna("").astype(str).str.strip().str.capitalize()
    df["Region"] = reg_clean.replace({"": "East", "East ": "East", "North ": "North", "South ": "South"})
    df["Order_Date"] = pd.to_datetime(df["Order_Date"])
    df["YearMonth"] = df["Order_Date"].dt.to_period("M").astype(str)
    return df

def plot_sales_trend(df):
    monthly = df.groupby("YearMonth").agg(Total_Sales=("Sales", "sum"), Total_Profit=("Profit", "sum"))
    fig, ax = plt.subplots(figsize=(12, 6), dpi=300)
    ax.plot(monthly["YearMonth"], monthly["Total_Sales"], marker="o", color="#1F4E79", linewidth=2.5, label="Total Sales")
    ax.plot(monthly["YearMonth"], monthly["Total_Profit"], marker="s", color="#2CA02C", linewidth=2.5, label="Total Profit")
    ax.fill_between(monthly["YearMonth"], monthly["Total_Profit"], color="#2CA02C", alpha=0.1)
    ax.set_title("Monthly Sales & Profit Performance Trend (2023 - 2024)", fontsize=14, fontweight="bold")
    ax.set_xlabel("Trading Month (Year-Month)", fontsize=11, fontweight="bold", labelpad=10)
```

```

ax.set_ylabel("Amount in USD ($) ", fontsize=11, fontweight="bold", labelpad=10)
ax.yaxis.set_major_formatter(ticker.FuncFormatter(lambda x, p: f"${x * 1e-3:.0f}K"))
ax.set_ylim(0, max(monthly["Total_Sales"])+100)
plt.xticks(rotation=45, ha="right", fontsize=9)
peak_idx = monthly["Total_Sales"].idxmax()
ax.annotate(f"Peak Month: {monthly.loc[peak_idx, 'YearMonth']}\n${monthly.loc[peak_idx, 'Total_Sales']}K",
            xy=(peak_idx, monthly.loc[peak_idx, 'Total_Sales']), xytext=(peak_idx - 3, monthly.loc[peak_idx, 'Total_Sales'] * 1.05),
            arrowprops=dict(facecolor="#D9534F", edgecolor="#D9534F", arrowstyle="→", lw=1.5),
            fontsize=9, fontweight="bold", color="#A93226", bbox=dict(boxstyle="round,pad=0.3", fc="white", lw=1.5))
ax.legend(loc="upper left", frameon=True)
plt.tight_layout()
fig.savefig(os.path.join(OUTPUT_DIR, "01_sales_trend.png"), dpi=300)
plt.close(fig)

def plot_top_5_products(df):
    top5 = df.groupby(["Product", "Category"]).agg(Total_Sales=("Sales", "sum"), Total_Profit=("Profit", "sum"))
    fig, ax = plt.subplots(figsize=(10, 6), dpi=300)
    bars = ax.barh(top5["Product"], top5["Total_Sales"], color=["#5DADE2", "#48C9B0", "#F5B041", "#5499C7", "#A93226"])
    ax.set_title("Top 5 Revenue-Generating Products", fontsize=14, fontweight="bold", pad=15, color="#A93226")
    ax.xaxis.set_major_formatter(ticker.FuncFormatter(lambda x, p: f"${x * 1e-3:.0f}K"))
    for bar, (_, row) in zip(bars, top5.iterrows()):
        w = bar.get_width()
        m = (row["Total_Profit"] / row["Total_Sales"]) * 100
        ax.text(w + 1000, bar.get_y() + bar.get_height() / 2, f"${w:.0f} (Margin: {m:.1f}%)", va="center", color="black", fontweight="bold", fontstyle="italic", size=10)
    ax.set_xlim(0, max(top5["Total_Sales"]) * 1.25)
    plt.tight_layout()
    fig.savefig(os.path.join(OUTPUT_DIR, "02_top_5_products.png"), dpi=300)
    plt.close(fig)

def plot_category_donut(df):
    cat_summary = df.groupby("Category").agg(Total_Sales=("Sales", "sum"), Total_Profit=("Profit", "sum"))
    total_sales = cat_summary["Total_Sales"].sum()
    cat_summary["Margin"] = (cat_summary["Total_Profit"] / cat_summary["Total_Sales"]) * 100
    fig, ax = plt.subplots(figsize=(8, 8), dpi=300)
    wedges, texts, autotexts = ax.pie(cat_summary["Total_Sales"], labels=cat_summary["Category"], autopct='%1.1f%%',
                                     colors=["#1F4E79", "#2E75B6", "#5DADE2", "#A9CCE3"][:len(cat_summary["Category"])],
                                     wedgeprops=dict(width=0.45, edgecolor="white", linewidth=2), textprops=dict(color="white", fontweight="bold", fontstyle="italic", size=10))
    for at in autotexts: at.set_color("white"); at.set_fontweight("bold")
    ax.text(0, 0.08, "Total Sales", ha="center", va="center", fontsize=11, color="#7F8C8D", fontweight="bold", fontstyle="italic", size=10)
    ax.text(0, -0.08, f"${total_sales:.0f}K", ha="center", va="center", fontsize=15, fontweight="bold", fontstyle="italic", size=10)
    labels = [f'{r["Category"]}: ${r["Total_Sales"]:.0f}K ({r["Margin"]:.1f}%)' for _, r in cat_summary.iterrows()]
    ax.legend(wedges, labels, title="Category Sales & Margins", loc="lower center", bbox_to_anchor=(0.5, -0.1), fontweight="bold", fontstyle="italic", size=10)
    ax.set_title("Category Revenue Contribution & Profit Margins", fontsize=14, fontweight="bold", pad=15, color="#A93226")
    plt.tight_layout()
    fig.savefig(os.path.join(OUTPUT_DIR, "03_category_sales_donut.png"), dpi=300)
    plt.close(fig)

def plot_regional_performance(df):
    reg = df.groupby("Region").agg(Total_Sales=("Sales", "sum"), Total_Profit=("Profit", "sum")).reset_index()
    fig, ax = plt.subplots(figsize=(10, 5.5), dpi=300)
    x = range(len(reg))
    w = 0.35
    b1 = ax.bar([i - w/2 for i in x], reg["Total_Sales"], w, label="Sales ($)", color="#1F4E79", edgecolor="black", fontweight="bold", fontstyle="italic", size=10)
    b2 = ax.bar([i + w/2 for i in x], reg["Total_Profit"], w, label="Profit ($)", color="#2CA02C", edgecolor="black", fontweight="bold", fontstyle="italic", size=10)
    ax.set_title("Regional Commercial Breakdown: Sales vs. Net Profit", fontsize=14, fontweight="bold", pad=15, color="#A93226")
    ax.set_xticks(list(x)); ax.set_xticklabels(reg["Region"], fontsize=11, fontweight="bold", fontstyle="italic", size=10)
    ax.yaxis.set_major_formatter(ticker.FuncFormatter(lambda y, p: f"${y * 1e-3:.0f}K"))
    for b in b1: ax.text(b.get_x() + b.get_width()/2, b.get_height() + 1500, f"${b.get_height()*1e-3:.0f}K", va="bottom", color="black", fontweight="bold", fontstyle="italic", size=10)
    for b in b2: ax.text(b.get_x() + b.get_width()/2, b.get_height() + 1500, f"${b.get_height()*1e-3:.0f}K", va="bottom", color="black", fontweight="bold", fontstyle="italic", size=10)

```

```
ax.set_ylim(0, max(reg["Total_Sales"]) * 1.18); ax.legend(loc="upper right", frameon=True)
plt.tight_layout()
fig.savefig(os.path.join(OUTPUT_DIR, "04-regional_performance.png"), dpi=300)
plt.close(fig)

if __name__ == "__main__":
    df = load_data()
    plot_sales_trend(df)
    plot_top_5_products(df)
    plot_category_donut(df)
    plot_regional_performance(df)
    print("Visuals regenerated successfully.")
```

Department of Computer Science & Engineering (Our Data Analytics Specialization)Candidate: Gourav

## Appendix E: Raw Data Quality Defects Audit Log ( `DATA_QUALITY_NOTES.md` )

```
# Raw Data Quality Notes & Anomaly Audit
**Source Dataset**: data/raw_sales_data.csv (3,000 transaction records)

1. Exact Duplicate Transactions (10 Rows):
  - Fully duplicated order rows placed at intervals to model POS double-entry retry anomalies.
  - Cleansed via unique Order_ID deduplication in Excel and Python.

2. Missing (Blank) Values (10 Rows):
  - Order_ID: 100% complete (zero nulls).
  - Sales: 2 missing entries (re-derived via Quantity * Price * (1 - Discount)).
  - Discount: 2 missing entries (re-derived via ((Qty * Price) - Sales) / (Qty * Price)).
  - Profit: 2 missing entries (re-derived via Sales - (Qty * Cost)).
  - Category: 2 missing entries (imputed deterministically via Product catalog lookup).
  - Region: 2 missing entries (imputed via Customer territory history).

3. Inconsistent Category Formatting (10 Rows):
  - Casing issues: "technology", "office supplies", "electronics", "TECHNOLOGY".
  - Spacing issues: "Technology ", "Furniture ", "Office Supplies ", " Furniture".
  - Standardized via TRIM() and Title Casing.

4. Inconsistent Region Formatting (5 Rows):
  - Casing issues: "north", "east".
  - Spacing issues: "West ", "South ", " North".
  - Standardized via TRIM() and Title Casing.

Overall Integrity Audit: >98.8% of raw records structurally valid out-of-the-box.
Cleaned Dataset Output: 2,990 rows (sales_data_cleaning.csv).
```